Land subsidence: Environmental risk in housing markets in Mexico City*

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Abstract

We study the costs of and the housing market response to subsidence– the sinking of land areas due to groundwater over-extraction– in Mexico City. We propose an equilibrium model of the housing market that features housing re-development in the face of a dynamic environmental hazard that impacts home quality. Our model highlights that frictions affecting the capitalization of future risk exacerbate the costs of hazards by causing an over-supply of housing in high-risk areas. Guided by model-derived estimating equations, we exploit quasi-random variation in sinking intensity to estimate the impact of both realized and future subsidence on home values and redevelopment. Realized subsidence imposes substantial costs, lowering home values by 1.15% on average. However, prices are unresponsive to measures of expected future sinking, and survey evidence suggests that information frictions affect the ability of homebuyers to capitalize predictable future risk. Units that have experienced more sinking are *more* likely to be redeveloped, as sinking acts as accelerated depreciation. Evaluating welfare implies that current rates of subsidence cost Mexico City a total of $8.27 billion USD, $3.96 billion of which are due to pricing frictions that inefficiently increase the housing stock in the riskiest neighborhoods.

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1 Introduction

As groundwater levels decline in places of high population density worldwide, policymakers are increasingly concerned about the environmental and economic consequences of falling water tables (Carleton et al., 2024; Rodella et al., 2023). One important yet under-studied externality resulting from groundwater decline is land subsidence, the sinking of the Earth's surface due to groundwater depletion. Land subsidence occurs when aquifer levels within a groundwater basin fall, leading to the irreversible loss of elevation for the overlying land, along with any infrastructure located in the area.[1] Subsidence is an externality because aquifers are hydraulically connected: a drop in water levels caused by pumping in one location eventually equalizes over space, causing land to sink in potentially distant areas within the same basin.

Land subsidence is a widespread and severe global problem. Sinking hotspots have been identified on almost every continent, and fast-sinking cities overlying severely over-drafted aquifers such as Jakarta, Tehran, and Mexico City can lose 20–30 centimeters of elevation per year.[2] While this represents relatively slow movement, the structural issues arising as buildings sink with the moving ground are acute. Sinking buildings experience cracking, tilting, water infiltration, and eventually total loss of structural integrity. Sinking public infrastructure such as water mains and roads also fissure and crack under the pressure of shifting soils, especially when this movement is uneven.

We study the costs of subsidence and the equilibrium housing market response to this evolving hazard in Mexico City. Mexico City is one of the largest and fastest-sinking urban

[1] Subsidence can also be induced by intensive mining activities or volcanic eruptions; in this paper we focus on groundwater depletion as this is the predominant cause in cities.

[2] There are particularly fast-sinking areas in coastal East Asia, Iran, Mexico, and the United States with areas of intense sinking in 45 states in the United States and more than 95 cities in China (Bagheri-Gavkosh et al., 2021). As Figure A1 illustrates, this issue tends to be particularly acute in developing countries with water management challenges. In subsiding coastal areas, it is estimated that up to 30% of projected sea level rise is in fact due to these areas sinking into the sea (Nicholls et al., 2021), and currently 11 coastal cities are at risk of becoming totally inundated by the end of the century. In a striking example, Indonesia has developed a plan to move the capital city away from Jakarta, which is rapidly subsiding below sea level due to groundwater extraction (Associated Press, 2022).

areas in the world, with the worst-hit parts of the city losing approximately a meter of elevation every 3-4 years (Chaussard et al., 2021). Figure 1 depicts some examples of what the consequences of subsidence look like in the city. Buildings on sinking land can become tilted and cracked or even begin emerging out of the sinking ground. Fissures and potholes in roads are common in sinking areas, and new patterns of flooding have emerged as slopes change in the city.

Estimating the costs of dynamic hazards like subsidence is complicated by the fact that forward-looking housing markets internalize both *realized* and expected *future* impacts to homes. This pricing of future risk may be imperfect, however; a large literature has observed that capitalization of environmental risks often lags behind predictable impacts in the absence of sudden and extreme events that correct the market. We formalize these dynamics and structure our approach to estimating the costs of subsidence by proposing a spatial equilibrium model of the housing market. Our model incorporates three key features of the problem that are general to many environmental risks: realizations of subsidence damage homes; there may be frictions affecting the capitalization of future damages; and re-development can 'reset' housing quality by erasing past realizations.

Our framework highlights the nuanced role that environmental hazards play in housing market dynamics. While expected future impacts tend to lower the equilibrium housing stock in risky areas, frictions affecting the pricing of expected future damages lead to an inefficiently high housing stock being built in harm's way. Realizations of the hazard operate through a different mechanism, as sinking realizations increase the rate of re-development by accelerating the depreciation of structures.

The model elucidates that three parameters are central to this relationship: the willingness to pay of residents to avoid subsidence, the extent of pricing frictions, and the price elasticity of housing supply. We derive theory-consistent, reduced-form estimating equations for these parameters from the model structure, and estimate them using novel measures of land subsidence, administrative data on the housing market, and an original

survey on potential sources of pricing frictions.

We measure subsidence by combining original processing of radio wave-based satellite products with pre-existing geophysical data, resulting in the longest, highest-resolution panel of sinking in Mexico City to date. We do so using a process called Synthetic Aperture Radar interferometry, a tool used by geophysicists to measure distortions in the Earth's surface with applications related to subsidence, volcanic eruptions, and glacier movement, among others. This new dataset measures annual sinking rates in Mexico City from 2007-2024 at 100-meter resolution, and allows us to map sinking trajectories onto individual properties in Mexico City.

With these measures in hand we take our model to the data, using the universe of mortgage appraisals in Mexico City to estimate the willingness to pay of home buyers to avoid subsidence and the capitalization of expected future subsidence into home values. In order to causally identify these parameters, we implement a repeat transaction design that exploits the fact that sinking on a given plot is caused by non-local groundwater pumping. We find that realized sinking is costly. Every meter of sinking causes a loss of 4.6% in appraised value, and the average home in our sample loses 1.15% of its value to sinking every 10 years. Evidence from a survey of long-term residents and our appraisal data suggests that losses are a product of both physical impacts to the home and increased maintenance costs.

In contrast to the large negative impact of realized sinking to values, expected future sinking is unpredictive of prices, and evidence from a novel survey in Mexico City sheds light on the mechanisms driving this lack of capitalization. We find evidence for substantial information frictions affecting homebuyers' decisions. Surveyed residents are unlikely use ex-ante available information to infer future sinking risk. Furthermore, we observe sorting on residents' ability to price future sinking, with residents of high-sinking plots less likely to use existing information about risk and more likely to report surprise about the severity of sinking. This is evident in our appraisal data: While past sinking is valued evenly across the city, expected future sinking is only capitalized in areas of *lower* sinking intensity. We also

show that discount rates and risk preferences are unlikely explanations for our results.

In our last estimation step we estimate the supply elasticity of re-development using a survey of housing developers. We infer that a property was re-developed by matching appraisals to the locations of new builds, and identify the supply elasticity by implementing a control function approach in which we instrument for prices using the unevenness of sinking across neighborhoods. Consistent with our model intuition, we find that a meter of sinking increases the probability that a plot is redeveloped by 3.9 percentage points. Interpreting our estimates structurally provides an estimate of the supply elasticity of 0.28.

Finally, we use our estimated model parameters to evaluate welfare under counterfactuals that enable us to quantify the total costs of subsidence and decompose these costs into the direct cost sinking and those arising from pricing frictions. We estimate that current rates of sinking in Mexico City impose a permanent cost of $8.27 billion US dollars, which represents 4.4% of the total value of housing in Mexico City. Back-of-the-envelope cost-benefit exercises suggest that several feasible policies promoting aquifer recharge, such as injecting wastewater into the aquifer or investing in alternative surface sources of water, would produce net benefits in this context.

In a counterfactual that isolates the role of pricing frictions, we find that pricing frictions are very costly and lead to important distortions in the geography of housing. We estimate that pricing frictions are responsible for about half of the total costs of subsidence, and that fully alleviating them reduces the housing stock in the highest-risk areas of the city by up to 2.5%. This finding underscores how behavioral or policy frictions that affect the efficient pricing of risk can severely amplify the costs of environmental hazards.

These findings represent the first comprehensive economic analysis of subsidence. While a large body of work has focused on the costs of environmental externalities such as air pollution, floods, and wildfires, subsidence is a natural hazard that has been relatively understudied by economists despite the fact that it is an important groundwater externality

affecting many major cities around the world.[3] We contribute to the sparse economics literature that has studied subsidence (Willemsen et al., 2020; Yoo and Perrings, 2017) by combining improved causal inference with market valuation to estimate costs, and by providing empirical and theoretical analysis of the equilibrium housing market response to sinking. Our original satellite work and survey contribute novel evidence on the physical impacts of sinking on buildings and public infrastructure, which to our knowledge is the first representative measure of damages for a city.

By modeling the housing market response to environmental risk in equilibrium, this paper is related to the literature on urban development and climate adaptation (Balboni, 2025; Desmet et al., 2021; Hsiang et al., 2017; Hsiao, 2023). Our model contributes to these papers by modeling housing redevelopment in the context of these hazards, which allows us to carefully examine the different roles that *realizations* and *expectations* of environmental risks play in driving housing supply and re-development. In this sense, this paper is closest to Hsiao (2023) who studies moral hazard in development in the context of sea level rise in Jakarta. Relative to this paper, we study how demand frictions contribute to an inefficient allocation of housing over space, and model re-development as a mechanism by which the housing stock is renewed after realizations of disasters.

Finally, our framework provides a way of quantifying the aggregate welfare costs of pricing frictions or imperfect information in housing markets. We build on recent empirical findings from work in environmental, urban, and real estate economics that study how information frictions about environmental risk impact housing markets (Bakkensen and Barrage, 2022; Bakkensen et al., 2019; Gallagher, 2014; Gourevitch et al., 2023; Hino and Burke, 2021; Holtermans et al., 2024; Lee and Zheng, 2023; Ortega et al., 2025; Ortega and Taşpınar, 2018; Petkov and Ortega, 2025; Wagner, 2022), and go further by embedding these frictions

[3]Engineering-style estimates of the costs of subsidence suggest that these may be large, especially in urban areas; case studies from China (Lixin et al., 2010), Indonesia (Mahya et al., 2021), Mexico (Fernández-Torres et al., 2025; Novelo-Casanova et al., 2022), and California (Borchers and Carpenter, 2014; Fowler, 1981) find damages in the billions for specific cities or counties. However, the general approach in these studies is to estimate replacement costs for damaged infrastructure without considering market values or forces.

in an equilibrium spatial model of the housing market to quantify the aggregate costs of these frictions.[4] Furthermore, our work provides rich empirical evidence on the housing market response to an important externality in a middle-income context, a setting where the necessary housing value data is rarely available.[5]

The rest of this paper is organized as follows. Section 2 introduces the context and our satellite data for measuring subsidence. Section 3 presents the theoretical framework. Section 4 presents the economic data we use, and Section 5 presents the estimation strategy and results. Section 6 presents model counterfactuals and policy analysis. Section 7 concludes.

2 Subsidence in Mexico City

Mexico City has experienced severe rates of subsidence over the last 100 years, which geophysicists have established is caused by sustained groundwater pumping on the Mexico Valley aquifer (Cabral-Cano et al., 2008; Chaussard et al., 2021; Solano-Rojas et al., 2015).[6] This pumping is primarily conducted by the city government and the federal water authority for the purpose of supplying the city with about two-thirds of its total water consumption, the other third of which is supplied by surface water sources.[7] Because the aquifer is hydraulically connected, pumping anywhere affects the water table everywhere (Cigna and Tapete, 2021), and the city itself experiences high rates of subsidence despite the fact that there is almost

[4]Our forward-looking demand framework also draws on insights from recent dynamic hedonic approaches (Rosen, 1974; Benetton et al., 2025; Bishop and Murphy, 2019), and we propose a way of incorporating information frictions into these methods.

[5]One important exception is Gonzalez-Navarro and Quintana-Domeque (2016), who also use professional appraiser data to study road paving in Mexico.

[6]Mexico City has a long history of water management and scarcity; the Aztec empire had built their capital city around and interwoven with the large lake system in the area, but the lake was subsequently drained by the Spanish colonial government. Because such a large part of the city is built on the soft, silty lake bed of the drained lake, this area is particularly prone to subsidence.

[7]Bringing surface water into Mexico City is very expensive due to how far large surface water sources are and the fact that most sources need to be pumped over the surrounding mountain ranges. The most important surface water source for the city is the Cutzamala system, which involves seven dams and over 330 kilometers of canals and pipes that use over 2 billion kWh of electricity per year to transport water to Mexico City.

no groundwater pumping conducted within city limits.[8]

Thus sinking rates are primarily a product of two factors: the height of the water table and the propensity of overlying geologies to compact and sink. Given that the water table is currently far below historic levels, sinking accelerates when the water table is relatively lower (and vice versa), and could only be fully stopped if the aquifer is recharged either naturally or via water injection (Gambolati and Teatini, 2021; Liu et al., 2019). While it is conceivable that the weight of human development could increase sinking, empirical studies in the Mexico City area have found no correlation between localized sinking rates and population density, pumping rates, or land use type (Chaussard et al., 2021).[9] Our own analysis in Appendix A1.1 supports this conclusion, as we find that construction on a pixel does not predict subsequent sinking rates.

This groundwater-induced subsidence can have severe consequences for building quality. Sinking homes experience cracking, tilting, water infiltration, and levelling issues arising from the total accumulated movement of the structure. Studies such as Willemsen et al. (2020) have found that total subsidence, rather than differential (spatially uneven) movement, is the principal factor affecting home quality, since substantial but relatively uniform movement across a plot still produces distortions arising from small variations in the underlying geology. While using pylons for the foundation when building can mitigate the impacts of sinking on a structure, these are rare in residential properties in Mexico City, and only 3.3% of respondents in high-sinking zones in our survey report that their foundation uses pylons.[10]

[8]The government has limited all pumping on the aquifer to the periphery of the city– a policy decision made after an episode of intense sinking in the 1950s (SACMEX, 2012). Displacing pumping alleviated sinking in the short-term because water table levels take substantial time to equalize over space. Pumped water is first sent to treatment facilities then to distribution stations, making the relationship where water is pumped and supply of water to a specific area tenuous.

[9]Studies have found that the weight of New York City has caused sinking of a couple millimeters, but New York has much higher population density than Mexico City, and this effect is orders of magnitude smaller than the rates observed in Mexico (Parsons et al., 2023).

[10]Pylons are much more common and even mandated by the construction code for large apartment and commercial buildings. However, these investments are rarely implemented in most residential buildings because the cost of doing so is too high relative to the low height and value of these structures.

2.1 Measuring subsidence

We measure sinking at fine geographic scale using a remote sensing technique borrowed from the Earth sciences called Synthetic Aperture Radar (SAR) interferometry. SAR instruments on satellites emit energy at the Earth in the form of radio waves and record a transformation of the number of oscillations that the wave undergoes between the earth and the satellite. While these measurements are meaningless on their own, constructing the difference between two images (interferometry) creates high-accuracy measures of ground displacement.[11] Several studies have empirically and mathematically characterized the measurement error associated with SAR measurements, and inaccuracies tend to run from 1 to 10 millimeters for modern satellites applied in urban areas (Wu and Madson, 2024), an order of magnitude smaller than the subsidence rates we measure in Mexico City. Appendix A1.1.1 discusses measurement error in detail.[12]

We apply established algorithms for conducting interferometry and subsidence time series construction that correct for distortions caused by terrain, tropospheric factors, and orbital error to construct an 18-year panel of subsidence rates at the 100-meter pixel level for Mexico City (Fattahi et al., 2017; Yunjun et al., 2019; Yu et al., 2018b). We do this by combining three series of data. We perform two rounds of original interferometry using data produced by the ALOS-1 (2007-2011) and Sentinel-1 (2021-2024) satellites. We combine these with pre-existing measures covering 2014-2020 that were created by geophysicists using Sentinel-1 readings (Chaussard et al., 2021). We linearly interpolate the data from 2011-2013, as no

[11] These calculations are deterministic in nature, and are not the result of machine learning algorithms like those well known in applications such as remote sensing of pollution or applications that use image-based sensors.

[12] Interferometry achieves higher accuracy in urban areas because the hard reflective surfaces minimize noise generated by signals intercepting vegetation. Construction does not have a strong impact on our measurements for two reasons. First, interferometry has a ceiling on the magnitude of elevation change it can detect between satellite passes, and no measure can be produced for changes exceeding the wavelength of the sensor (approximately 5 cm every 12 days in the case of Sentinel-1). Thus construction, which involves fast changes in building height, are generally not measurable. On a practical note, because we measure sinking on a property starting from the date construction was finished, any distortions in sinking resulting from construction would not affect our quantification exercises.

SAR measurements were taken for Mexico over this period.[13]

Finally, we define measures that characterize the gradient of sinking over space. Uneven sinking across larger spatial areas (i.e., between neighborhoods) is the principal risk factor for damage to public infrastructure such as distortions in roads and water mains or increased flood risk. To capture this larger scale variation, we define a 500-meter grid over Mexico City (the median census tract), and calculate the grid cell's *relative* displacement between periods t and t' as the absolute value of the accumulated difference in grid cell sinking and the average of its neighbors:[14]

$$\text{Abs. relative elevation change}_{gtt'} = |\sum_{\tau=t}^{t'}(s_{g\tau} - \overline{s}_{g\tau}^{nbrs})| \tag{1}$$

Because this measure accumulates over time, it allows us to capture the cumulative impact of relative altitude change in the environment as roads and water mains experience increasing strain.

2.2 Sinking is fast and predictable in Mexico City

Our subsidence data paint a detailed picture of sinking patterns in Mexico City in recent decades. Figure 2, Panel (a) plots sinking rates across Mexico City in 2010. The highest rates of subsidence across the city are around 20–25 centimeters per year, rates that are among the highest in the world (Werner et al., 2013). These fast rates are strongly localized in the eastern part of the city, which overlies particularly vulnerable geologies prone to subsidence.[15] Panel (b) plots the relative altitude change, or the unevenness sinking. Highly uneven sinking is concentrated at the edges of the high-sinking area, as well as around

[13]By using three separate series, we are able to address the computational challenge presented by calculating a single time series. The raw Single-Look Complex (SLC) SAR data and its processed output for just one orbit direction (ascending) for 5 years contains almost five terabytes of data and takes over three weeks of continuous computational time to process.

[14]This measure takes inspiration from the horizontal gradient used in geophysics studies such as Fernández-Torres et al. (2020).

[15]Geophysicists estimate that this part of the city could sink up to 30 more meters given the thickness of the features of the subterranean geology prone to compaction (Chaussard et al., 2021).

relatively fixed geologic features that maintain their altitude as the neighborhoods around them sink.

While sinking intensity varies year to year due to groundwater conditions, droughts, and seismic events, subsidence on a broad scale is very predictable over space due to the concentrated geological features that determine risk.[16] Figure 3 illustrates the spatial predictability of sinking by plotting the R^2 of regressions that predict the sinking rate on a pixel using sequentially more granular spatial fixed effects, including and excluding the lagged rate.[17] Even without the lagged rate, knowing the 5-kilometer grid cell a plot is located on already explains over half of the variation in sinking, and knowing the pixel location explains almost 80% of the variation. If the lagged sinking rate is included, this predictive power goes up to 83%. Perhaps because of this broad systematic predictability, subsidence is largely uninsurable in Mexico City, and is explicitly listed in home insurance policies as a non-covered event.[18]

3 Theoretical framework

Subsidence, as with other environmental hazards such as floods, earthquakes and wildfires, has both *realized* damages and a systematic risk profile over space which generates *anticipated* impacts to homes. We incorporate these dynamics into a model of the housing market that features forward-looking demand and re-development that replaces damaged units with yet-unsunk ones. Given the low prevalence of adaptive investments in residential buildings in Mexico City, we model re-development as a mechanism through which the housing stock is restored (replaced), but assume re-development has no adaptive benefits. Our model yields predictions about how realized and expected sinking affect the market, welfare expressions

[16] Appendix A2 provides detailed analysis of the data generating process of subsidence and its predictability.

[17] We plot the unadjusted R^2 to imitate an "out-of-sample" prediction exercise of adding another year of data conditional on location as opposed to an out-of-sample that conceptualizes adding another spatial unit.

[18] Subsidence was recently included in "home quality insurance" policies that cover INFONAVIT mortgages from the federal government, but these policies only cover the first 10 years of the home's life, within which the most severe impacts of sinking would not yet be observed.

that allow us to quantify the costs of subsidence, and estimating equations for the key parameters needed to conduct welfare analysis.

Set-up Time is discrete and indexed by periods t. Mexico City is partitioned into a discrete set of plots indexed by j, and $j = 0$ represents locations outside of Mexico City. Each period begins with a continuous measure of housing on each plot j in Mexico City, H_{jt-1}, with the initial measure of housing H_{j0} exogenously given. All units on a given plot are identical. The following sequence occurs in each period:

1. Exogenous subsidence shocks $s_{jt} \sim \mathcal{S}_j$ are realized on each plot.

2. The developer draws cost shocks for developing each plot and all housing units go up for sale on the market to own housing.[19] Units may be purchased by another homeowner or the developer. If they are purchased by the developer, they demolish the existing units and build new ones, which have zero realized sinking and re-enter the market for purchase by homeowners. The ownership market clears on the price of owning a home p_{jt}, and each plot has a housing stock H_{jt} that is fixed until the next period.

3. Homeowners supply rentals from the housing they purchased in the previous step. Renters draw preference shocks over plots and make a discrete choice over which plot to live on. Equilibrium rents R_{jt} adjust to clear the rental market.

3.1 Housing demand

Demand for sinking homes is the outcome of a rental market equilibrium in which all housing is occupied by renters who rent from homeowners.[20] Because homeowners purchase a durable asset, they must form expectations over future sinking, which may be biased.

[19] By putting the full stock on the market in each period, we abstract from housing inventory and matching buyers to available housing.

[20] This separation is useful because it allows us to separate the flow utility provided by the home (which accrues to renters) from the asset gains or losses (which accrue to homeowners), but note that the case of owner-occupiers (65% of residents in Mexico City) is nested in the case of homeowners who rent to themselves.

3.1.1 Renters

Our model of renters captures the disutility from living in a home that has already sunk, as these housing units may be cracked, tilting, have water leaks, or other issues that occupants either dislike directly or because remedying these issues involves hassle and monetary costs.

A continuum of renters in Mexico indexed by i with mass $\overline{L}$ draw preference shocks over plots and decide which plot to rent a unit of housing on. Renters are perfectly and costlessly mobile, and derive utility/disutility from paying rent R_{jt}, how much the unit has sunk to date, S_{jt}, other observable attributes Z_{jt} and unobserved attributes e_{jt}, each of which are unaffected by subsidence. Total subsidence S_{jt} is the accumulation of sinking rates on the plot $s_{j\tau}$ from the time of construction to t:

$$S_{jt} = \sum_{\tau=\text{Year built}_j}^{t} s_{j\tau}$$

We assume renter utility is given by:

$$V_{jt}^{i} = -\log(R_{jt}) + \gamma S_{jt} + Z_{jt}\delta + e_{jt} + \epsilon_{jt}^{i} \tag{2}$$

where $\epsilon_{jt}^{i} \sim$ iid EV(1) with variance 1. We normalize mean utility from living outside Mexico City to zero: $V_{0t}^{i} = \epsilon_{jt}^{i}$.

These assumptions imply that the probability that a renter chooses plot j is:

$$\lambda_{jt} = \frac{\exp(-\log(R_{jt}) + \gamma S_{jt} + Z_{jt}\delta + e_{jt})}{1 + \sum_{l} \exp(-\log(R_{lt}) + \gamma S_{lt} + Z_{lt}\delta + e_{lt})} \tag{3}$$

and total rental demand is given by:

$$H_{jt}^{D} = \overline{L}\lambda_{jt} \qquad \forall\, j \tag{4}$$

3.1.2 Homeowners

Homeowners act as firms who purchase housing in anticipation of a future stream of rental income. These entities thus internalize the value of expected future sinking, the mechanism that generates forward-looking demand.

In each period a mass of homeowners, indexed by $k \in \Omega$, purchase housing (either used or newly developed) and rent units to renters. Their expected profits from holding a measure of housing $H_{jt}(k)$ are given by:

$$\tilde{\mathbb{E}}_t(\pi_{jt}(k)) = H_{jt}(k)\tilde{\mathbb{E}}_t\left(\sum_{\tau=t}^{\infty} \rho^{\tau-t} R_{j\tau}\right) - p_{jt}H_{jt}(k) \tag{5}$$

Expected revenue is the present value of expected rents accruing from renting $H_{jt}(k)$ of the housing measure. Homeowners form subjective expectations $\tilde{\mathbb{E}}_t()$, which may differ from rational expectations $\mathbb{E}_t()$. Each unit of $H_{jt}(k)$ has an up-front cost of p_{jt} to purchase from another homeowner or the developer.[21] Their total expected profits sum over their holdings on all plots:

$$\tilde{\mathbb{E}}_t(\pi_t(k)) = \sum_j \tilde{\mathbb{E}}_t(\pi_{jt}(k))$$

3.1.3 Frictions affecting evaluation of future risk

Expectations over future sinking play an important role in (5). However, frictions such as myopia, information frictions, or distortionary policies may complicate the valuation of yet-unrealized damages.[22] We incorporate the potential for systematic mis-valuation of future risk by modeling expectations of future sinking rates as a function of rational expectations

[21] We abstract from leverage or other financial markets by assuming that they self-finance their capital investment into the property.

[22] This is especially true in the more realistic case of atomistic home buyers who cannot hedge over a sophisticated real estate portfolio (Bakkensen and Barrage, 2022; Gourevitch et al., 2023; Gallagher, 2014) and purchase a limited number of properties over their lifetime, limiting the scope for learning over time. The model can be extended to policies such as bailout programs or subsidized insurance schemes that also introduce a wedge between internalized and actual future risk if these policies can be reasonably modeled as a proportional (multiplicative) reduction in anticipated costs.

such that:

$$\tilde{\mathbb{E}}_t(s_{ju}) = \theta \mathbb{E}_t(s_{ju}), \qquad u > t \tag{6}$$

In this expression, θ parameterizes the extent of these frictions on forming expectations about future sinking. When $\theta = 1$, homebuyers have rational expectations about future sinking and fully internalize these expected future costs; as $\theta \to 0$, buyers shade their expectations about the impacts of future sinking.[23]

3.2 Housing supply

The supply side is designed to capture two margins along which the stock of housing responds to environmental hazards. First, development involves replacing existing, damaged structures with new ones that have not yet sunk. Second, supply is responsive to market prices, which internalize realized and future sinking.

We assume housing units in Mexico City are supplied by a housing developer who builds and renews the housing stock by purchasing existing housing units, demolishing them, and rebuilding. Purchasing a plot means purchasing the full measure of housing contained on it, H_{jt-1}. Newly built units have zero realized sinking, and we assume future sinking cannot be prevented. In each period, the developer receives plot-specific cost shocks $\xi_{jt} \sim \text{LogNormal}(\mu, \sigma^2)$ and decides which plots to purchase and redevelop in each period. Their plot-specific profits conditional on developing are:

$$\pi_{jt} = \max_{H_{jt}} \underbrace{p_{jt}(0)H_{jt} - \xi_{jt}H_{jt}^{\frac{1+\eta}{\eta}}}_{\text{Variable profits from developing}} - \underbrace{p_{jt}H_{jt-1}D}_{\substack{\text{Cost of acquiring,} \\ \text{demolishing existing units}}}, \qquad \eta > 0 \tag{7}$$

[23] This behaves similarly to the β in the beta-delta behavioral model (Laibson, 1997; DellaVigna, 2018), except that the discounting is only over future sinking and not all future rents. In this model, θ does not update with realized sinking, but the total welfare impacts of these frictions are the same in an alternative model in which $\theta < 1$ only for the period that the home is first sold as a new unit, which changes only which homeowners experience negative profits. We model that θ is uniform among homebuyers, but note insights from papers such as Bakkensen et al. (2019) that show that only some mis-informed participants are needed to inflate prices.

where η is the supply elasticity, and $p_{jt}(0)$ is shorthand for the price that a developer receives for new units that have experienced no sinking ($S_{jt} = 0$). D is cost of demolishing existing units, which is proportional to the total pre-development value of the plot.[24] Because the developer competes with homeowners (the used housing stock) both when selling their units and when purchasing plots, they take market prices as given but may make positive profits arising from favorable cost shocks.

Conditional on developing, the developer chooses how many units to build optimally. Solving this problem gives:

$$H_{jt}^{opt} = \left(\frac{\eta}{(1+\eta)\xi_{jt}} p_{jt}(0) \right)^{\eta}$$
$$\pi_{jt}^{*} = \max\{0,\ \tilde{\eta}\xi_{jt}^{-\eta} p_{jt}(0)^{\eta+1} - p_{jt} H_{jt-1} D\}, \qquad \tilde{\eta} = \left(\frac{\eta}{1+\eta}\right)^{\eta} \left(\frac{1}{1+\eta}\right)$$

The developer will re-develop a plot if profits are positive:

$$d_{jt} := 1\left(\pi_{jt}^{*} > 0\right) \tag{8}$$

Housing supply on plots $j > 0$ in Mexico City is therefore:

$$H_{jt}^{S} = d_{jt} H_{jt}^{opt} + (1 - d_{jt}) H_{jt-1} \qquad \forall\, j > 0 \tag{9}$$

and we assume housing units outside Mexico City are perfectly elastically supplied at price equal to 1.

3.3 Equilibrium and welfare

Given exogenous parameters $\{\gamma, \theta, \eta, \overline{L}, \{H_{j0}\}_j, \delta, \rho, D, \mu, \sigma\}$, the distributions of sinking shocks $\{\mathcal{S}_j\}_j$, realizations of cost shocks $\{\xi_{jt}\}_{j>0}$ and sinking rates $\{s_{j\tau}\}_{j,\tau \leq t}$, the equilibrium

[24] This allows demolition costs to increase with buildings that are built with better materials or that are larger, qualities that would be correlated with higher value.

consists of a set of prices $\{p^*_{jt}\}_j$, rents $\{R^*_{jt}\}_j$, and housing $\{H^*_{jt}\}_j$ that satisfy the following conditions in each period:

- Renters choose the plot to rent on optimally. Homeowners and the developer maximize profits.
- Homeowners make zero subjectively-expected profits:

$$p^*_{jt} = \tilde{\mathbb{E}}_t \left(\sum_{\tau=t}^{\infty} \rho^{\tau-t} R^*_{j\tau} \right) \qquad \forall\, j \tag{10}$$

- The rental market clears on rents R^*_{jt} in each period such that rental demand given by (4) equals (9).[25]
- The market for new homes clears on home prices p^*_{jt} according to (10) and (9) given the price function $\log(p^*_{jt}(0)) = \log(p^*_{jt}) - \gamma S_{jt}$.[26]

The total change in welfare in the housing and rental markets in Mexico City between counterfactuals corresponds to the total change in developer, homeowner, and renter surplus between alterative equilibria.[27] Per-period developer's surplus is their total expected profits across all plots, taking expectations over the distribution of cost shocks:

$$PS_t = \sum_{j>0} \mathbb{E}_\xi(\pi^*_{jt}) \tag{11}$$

Expected homeowners' profits are derived by substituting (5) into (10):

$$\Pi_t = \sum_j H^*_{jt} \left(\sum_{\tau=t}^{\infty} \rho^{\tau-t} \left(\mathbb{E}_t(R^*_{j\tau}) - \tilde{\mathbb{E}}_t(R^*_{j\tau}) \right) \right) \tag{12}$$

[25] Given that (i) development decisions and the housing stock on each plot are determined before the rental market occurs, and (ii) conditional on ownership, property managers supply rentals at zero marginal cost, rental supply is inelastically supplied according to (9).

[26] Because there is only one developer competing with many homebuyers to purchase units, the transaction price will be set by homebuyer demand. This also implies that the expectation that homebuyers form of the future stream of rents will be set by the value of those rents conditional on *not* redeveloping, as that is the price a developer will pay them for the asset even if the plot is subsequently redeveloped.

[27] Derivations of the welfare terms and comparative statics in the next section can be found in Appendix A4.4.

Under rational expectations, these are zero; in the presence of pricing frictions (rationally) expected profits will be negative.[28]

To account for using log rents in utility, we characterize the change in renter welfare by calculating the equivalent variation Δ_t^D that equalizes average consumer surplus between counterfactuals. The present value of the difference in total welfare between counterfactuals is:

$$\Delta W = \Delta \Pi_t + \frac{\Delta_t^D}{1-\rho} + \frac{\Delta PS_t}{1-\rho} \tag{13}$$

3.3.1 Model predictions

Examining how the housing market behaves in equilibrium sheds light on how subsidence and pricing frictions affect the location and dynamics of housing. We list the main propositions here, and provide proofs in Appendix A4.4. For each experiment, we consider the "long-run" difference between equilibria under different parameter values, allowing prices and the housing stock to fully adjust.

Proposition 1: Future sinking and the housing stock Expected future sinking on a plot decreases the equilibrium housing stock there.

Proposition 2: Pricing frictions and the housing stock Pricing frictions leading to under-valuation of future sinking ($\theta \to 0$) increase the expected housing stock on sinking plots.

Proposition 3: Realized sinking and the flow of development Realized sinking on a property increases the probability of re-development.

Intuitively, expected sinking acts as a typical urban disamenity, dissuading housing development in sinking areas in equilibrium. However, under-valuing future sinking leads to an inefficiently high price being offered for new units in sinking areas, which implies

[28] Behavioral biases that introduce wedges between rational and subjective expectations imply direct losses to homeowners. The model can be extended to bailout programs or other policy-induced wedges, in which case these social profits capture the fiscal value transferred to homeowners.

an oversupply of housing conditional on developing.[29] Realized sinking accelerates the depreciation of sinking plots, and thus draws in faster redevelopment by decreasing the continuation value of the existing units, p_{jt}.

In this model three parameters emerge as crucial for quantifying the total costs of subsidence: γ, the per-period direct costs of subsidence, θ, the extent of pricing frictions, and η, which governs the shape of the supply curve and thus the responsiveness of supply to the market distortion. In the next sections, we derive estimating equations from the model for these parameters and estimate them using our data on mortgage appraisals and development.

4 Data

4.1 Measuring home prices: Appraisals

We measure housing values using confidential data on the universe of bank appraisals in Mexico City from 2005-2020. These appraisals are conducted for the purpose of financing home sales by the Federal Mortgage Society (SHF, *Sociedad Hipotecaria Federal*), and were made available through freedom of information request.[30] We use property characteristics and exact locations to identify identical units from the same building, a matching process that results in 17.4% of our sample as being identified as a repeat appraisal.[31] Sinking

[29] To a first order pricing frictions have no effect on the probability of re-development in this model. Because pricing frictions inflate housing density in sinking areas over the long-run, both the developer's fixed cost of buying the plot (which is proportional to housing density) and revenue (a function of expected future sinking) are too high under pricing frictions. Because we assume prices are log separable in sinking to a first order approximation, these terms exactly cancel. Note however that this is sensitive to the assumption of how sinking enters prices.

[30] An SHF appraisal is required as a condition of obtaining a mortgage from a private bank or the public mortgage financier, and all SHF appraisals must be done by an SHF-certified appraiser. Thus, these data contain the universe of SHF appraisals, but do not include unofficial appraisals that may have been done for cash purchases of homes. The 2020 National Housing Survey suggests this restriction is important, as only 30% of home purchase in Mexico City in 2020 were financed with some form of formal credit, with the majority of homes obtained via self-financing of own construction.

[31] We exactly match appraisals on the built square footage, the square footage of the property, the number of rooms and bathrooms, number of floors, number of parking spaces, and whether the building has an elevator, and require the coordinates of the appraisal to imply a less than 100-meter distance between them.

histories and futures are assigned to appraisals by spatially matching appraisals to pixels, and we use the dates of construction and sale to characterize sinking trajectories.

Mortgage appraisals in our setting offer a comprehensive, and in some ways preferable, measure of housing valuation relative to transaction values. An advantage of using mortgage appraisals is that mortgaged sales represent relatively arms-length transactions, which are less likely to be affected by relational contracting or fraudulent reporting for tax evasion, two common issues affecting price data from other sources in Mexico (Gonzalez-Navarro and Quintana-Domeque, 2016). However, in order for appraisals to capture the true valuation of subsidence by homebuyers as opposed to the appraiser's algorithm, they must contain a strong signal of the transaction price ultimately paid. Conceptually, this holds because appraisals are used for the purpose of valuing a mortgage, and both appraisers and the buyer have a strong incentive to appraise the home at the price a buyer is willing to pay.

We validate this alignment between appraisal and transaction values using a confidential dataset of appraisal and transaction prices called the R04, whose access was provided by the *Banco de México*. The R04 data only contain approximate locations, but allow us to directly compare appraisal and transaction prices. We find that 73% of observations have identical appraisal and transaction values, and that appraisal values closely track transaction prices. Figure 4 plots transaction prices against appraisals, showing that points are closely centered around the 45° line.[32] Supplementary Table A2 reports coefficients and goodness-of-fit statistics for regressions of transaction prices on appraised values. We estimate a coefficient of determination of 0.97 (Column (1)), highlighting the close relationship between appraisals and transaction prices.

Finally, we show that absolute deviations in appraisal values from sales prices are not correlated with subsidence in Column (3) of Table A2. This is crucial, as even small variations

Supplementary Table A1 describes how repeat sales properties differ from single sales in the data; properties with repeat sales are much more likely to be apartment as opposed to single family homes, and are generally newer, smaller, and less valuable than the full sample.

[32]Figure A2 in the appendix reports the distribution of deviations between appraised and transaction values, showing the full distribution of differences.

between appraisals and transaction values could bias results if it were exactly subsidence generating the difference between them.

4.2 Housing development

We measure the supply side of the market using the DIME (*Dinámica del Mercado Inmobiliario*) dataset, which is a quarterly survey of new housing developers.[33] From this survey we construct a dataset of 20,380 housing developments from 2007-2020, which are new construction projects that comprise one or more housing units.[34] The data are skewed towards larger developments and exclude individuals who contract new builds or renovations: the median housing development included 20 new units, but some developments have hundreds of new units.

We use these data to characterize the relationship between re-development and subsidence in two ways. First, we identify appraisals of used homes that were subsequently re-developed using a combination of address and spatial matching.[35] Our resulting dataset implies that 0.45% of appraised transactions were for redevelopment, representing about 0.02% of the total housing stock in a given year. Because addresses are not always consistently recorded in the data and may change when a plot is redeveloped, we likely miss appraisals that subsequently became redeveloped. We address this in an alternative analysis that avoids matching on appraisals by directly characterizing the geography of new development at the pixel-year level.

[33] These data are compiled by a private real estate consulting firm and access was granted through *Banco de México*.

[34] Given that new homes comprise 70-80% of the housing market in Mexico and that real estate developers built 60% of new homes, the DIME data observes 42-48% of the housing market in a given quarter (Rodríguez Zamora, 2010). This is likely a lower bound for Mexico City, which has a more formal construction sector than other areas of the country. Appendix A1.4 discusses coverage in more detail.

[35] While construction firms do not take out traditional mortgages, SHF appraisals are also requisite for bridge loans commonly used to finance the upfront costs of a new development.

4.3 Original survey

We conducted an original survey of 800 Mexico City residents to investigate potential sources of pricing frictions when evaluating land subsidence and provide descriptive evidence on the damages and costs from sinking. The survey was designed to be representative of residents with a long-term stake in their home, and respondents were restricted to owner-occupiers (73%) or those who had lived in their home for at least five years (27%). The survey was stratified based on subsidence risk, with two-thirds of the sample drawn from the seven high-risk boroughs and one-third from the remaining nine boroughs.[36]

This first-of-its-kind survey was designed around two main modules. First, before mentioning subsidence, we surveyed residents on the physical quality of their home and neighborhood as well as maintenance expenditures and activities. In the second module, we then mentioned subsidence by name, surveying residents on their knowledge about subsidence, how they thought about subsidence before living in their current home, and implemented Falk instruments for measuring discount rates and risk aversion (Falk et al., 2016). We subsequently use the location of respondents' homes to map their property to our satellite measures of sinking and infer the total amount of sinking their home has experienced since construction and the unevenness measures associated with their grid cell.

5 Estimation

In this section we leverage our data on home values and redevelopment to causally estimate the main parameters governing model dynamics on the demand (γ, θ) and supply (η) sides of the model.

[36] More details about survey sample definition and representativity can be found in the Data Appendix A1.2.

5.1 The willingness to pay to avoid subsidence

Our granular data on home values permits a direct approach to estimating the willingness to pay to avoid subsidence and capitalization of future risk that leverages within-building variation to isolate plausibly exogenous variation in sinking.[37]

In order to arrive at an estimating equation, we solve Equation (3) for rents:

$$R_{jt} = \exp\{\gamma S_{jt} + \delta Z_{jt} + e_{jt} - \tilde{\lambda}_{jt}\}$$

where $\tilde{\lambda}_{jt} = \log(\lambda_{jt}) - \log(\lambda_{0t})$, and substitute this expression into (10) to derive a relationship between log prices and sinking:

$$\log(p_{jt}) = \gamma S_{jt} + \log\left(\exp(-\tilde{\lambda}_{jt} + Z_{jt}\delta + e_{jt}) + \tilde{\mathbb{E}}_t(\sum_{\tau=t+1}^{\infty} \rho^{\tau-t} \exp\{\gamma S_{j\tau}^{t+1} + Z_{j\tau}\delta + e_{j\tau} - \tilde{\lambda}_{j\tau}\})\right)$$

where $S_{j\tau}^{t+1} = \sum_{u=t+1}^{\tau} s_{ju}$ is the cumulative sinking from $t+1$ to τ.[38]

Stronger assumptions are required to make progress towards an estimating equation that teases expected sinking from the residual term. We first assume that $Z_{j\tau}$ are time-invariant and can be written as Z_j. Next, we take two first-order approximations: first of $\mathbb{E}_t(\exp(\gamma S_{j\tau}^{t+1} + e_{j\tau} - \log(\lambda_{j\tau}) + \log(\lambda_{0\tau})))$ around the expectations of these variables, and

[37] Our approach also allows us to recognize that sinking is an individual property attribute in our appraisals data. We also note the equivalence of hedonic and discrete choice estimators when the characteristic is continuously "supplied" in the market, as is the case of subsidence in our setting (Bayer et al., 2007).

[38] We are able to factor out realized sinking from the expression because (i) its impact on homes is persistent, and (ii) because the transaction price is set by homebuyer demand, expected future rents are those of the non-redeveloped property.

then of the log function around $\mathbb{E}_t(e_{j\tau}) = e_{jt}$, $\mathbb{E}_t(\tilde{\lambda}_{j\tau}) = \tilde{\lambda}_{jt}$, and $\mathbb{E}_t(S_{j\tau}^{t+1}) = 0$ to arrive at:[39]

$$\log(p_{jt}) \approx \gamma S_{jt} + \gamma\theta \sum_{\tau=t+1}^{\infty} \rho^{\tau-t}\mathbb{E}_t(s_{j\tau}) + \delta Z_j - \log(H_{jt}^S) + \log(H_{0t}^S) + u_{jt}, \tag{14}$$

$$u_{jt} = e_{jt} + (1-\rho)\sum_{\tau=t+1}^{\infty} \rho^{\tau-t}\left(\mathbb{E}_t(e_{j\tau}) - e_{jt} + \tilde{\lambda}_{jt} - \mathbb{E}_t(\tilde{\lambda}_{j\tau})\right) - \log(1-\rho)$$

and formulate the estimating equation we take to the data as:

$$\log(p_{jt}) = \gamma S_{jt} + \gamma\theta \sum_{\tau=t+1}^{\infty} \rho^{\tau-t}\mathbb{E}_t(s_{j\tau}) + \alpha_j + \alpha_{v(j)t} + \alpha_{z(j)t} + \tilde{u}_{jt} \tag{15}$$

By including fixed effects for identical units in the same building α_j, we flexibly control for both observed and unobserved static quality differences between appraised units (Z_j, e_j), a concern that is first-order in many hedonic estimation strategies. The α_j also absorb $\log(H_{jt}^S)$, as housing supply on the plot is fixed between appraisals. Vintage by year fixed effects $\alpha_{v(j)t}$ flexibly control for age and other ways that the value of buildings of different vintages may evolve over time, and borough by year fixed effects $\alpha_{z(j)t}$ absorb housing supply outside Mexico City ($\log(H_{0t}^S)$), and allow demand shocks to vary flexibly across housing markets.

5.1.1 Specifying expectations and estimating θ

Estimating (15) requires taking a stand on rational expectations $\mathbb{E}_t(s_{j\tau})$. We formulate expectations as both an AR(1) process estimated at the 500-meter grid cell level, and as a "perfect foresight" measure that uses observed sinking on the plot post-transaction plus a projection for post-2020 sinking.

We take a conservative approach to inference on θ given the inherent risk of misspecification, and the fact that our rich controls that facilitate identifying γ also

[39] Appendix A4.1 provides more details and derivations. The second approximation is as if homebuyers expect demand shocks and relative rental demand to follow a random walk. In the case of the $\tilde{\lambda}_{jt}$ this approximation comes at a cost in terms of model consistency, as these relative choice probabilities are equilibrium objects that have a distribution that is not likely to be a random walk. However, this approximation is chosen to reflect the fact that $\tilde{\lambda}_{j\tau}$ is fixed until re-development; because re-development is rare, with a per-period probability of 2.7%, $\mathbb{E}_t(\tilde{\lambda}_{j\tau}) = Pr(d_{jt}=0)\lambda_{jt} + Pr(d_{jt}=1)\mathbb{E}_t(\tilde{\lambda}_{j\tau}|d_{jt}=1) \approx \lambda_{jt}$.

absorb much of the expectation-generating information that buyers may use to infer risk (Dickstein and Morales, 2018). We explore capitalization in depth in Section 5.2, where we implement alternative tests that relax the controls used in Equation (14) and focus on new builds. This approach allows more systematic information to enter the price function, but ultimately provides only suggestive evidence as it relies on a selection on observables assumption to estimate θ.[40] Given this, we complement our findings with detailed evidence from our survey, and simulate welfare impacts under different levels of pricing frictions.

5.1.2 Identification

Identification of the structural parameters in Equation (15) requires that the expected amenities and rental demand in the residual be independent of sinking and expected sinking conditional on the controls. Intuitively, our repeat-transaction approach exploits the intersection of three sources of variation in realized and expected sinking: changes in sinking intensity over time within a plot, variation across plots in sinking intensity, and the length of time between transactions. The use of property-level fixed effects break up the strong cross-sectional correlation in both sinking and prices, as can be seen in a map of residualized sinking and prices in Supplementary Figure A4. However, shocks to local housing markets may nonetheless be correlated with sinking intensity.

We first test for this by using new builds in our sample to test for a relationship between the composition of the housing market and subsidence. While our approach directly controls for quality, a correlation of the characteristics of units on the market with subsidence could be an indication that the market is evolving in other unobserved ways that confound our estimates. Supplementary Figure A3 reports the coefficients and 95% confidence intervals from regressions of the Z-scores of unit amenities on expected future sinking on the property. Sinking is uncorrelated with these amenities in most cases, and all point estimates imply a treatment effect less than 0.35 standard deviations per meter of expected sinking. For the

[40] Limiting to new builds is useful because these have no realized subsidence, which may serve as a sufficient statistic for future sinking.

hedonic index that aggregates all of these, the (insignificant) treatment effect of a meter of subsidence is less than 0.01 standard deviations.

We also examine the relationship between housing market trends and subsidence using the microdata from the Income and Expenditure Survey (ENIGH) to test whether future sinking predicts current rents, which need not be forward-looking. Supplementary Table A3 reports results from estimating the impact of future and realized sinking on log rents conditional on hedonic controls and locality-level fixed effects. Realized sinking has a noisy but a negative and largely economically meaningful impact on rents. By contrast, future sinking has no predictive power for log rents, and we can reject that each meter of future sinking is associated with impact on future rents larger than 1%. In all estimates the impact of future sinking on rents is an order of magnitude smaller than that of realized sinking. While these estimates suffer from low power and limited ability to control for unobservables, they provide reassurance that property values are not evolving systematically before sinking occurs.

Section 5.4.1 presents additional pre-trends and placebo tests and other robustness exercises, but we now proceed with the presentation of our results estimating the costs of subsidence.

5.1.3 Results: The direct costs of sinking, γ

Table 1 reports the results of estimating Equation (15). Regardless of the manner of specifying expectations or the discount rate chosen, realized sinking has a consistent and negative impact of appraised values, with a meter of sinking causing a 4.6% fall in appraised value. Given that the average property in our sample sank 25 centimeters over 10 years, this implies a 1.15% average decrease in values per decade. In these specifications we do not detect an economically or statistically meaningful effect of future sinking on prices, suggesting that homebuyers do not price changes in expected sinking conditional on static property characteristics.

5.1.4 Mechanisms: How does realized sinking affect homes?

What drives the direct costs of realized sinking? Residents may dislike living in sunk homes because they are damaged, or because they must invest time and resources in repairing them.

We use additional information from appraisals and our survey to explore whether sinking damages homes in Table 2. We first test whether sinking causes degradation in the structural quality of the unit in Column (1), which reports the result of estimating our repeat appraisal specification using the log appraised years of life remaining as the outcome. This measure captures the appraiser's estimate of the years of inhabitable life left for the structure, and provides information on how degraded or depreciated the physical building is. We find that sinking has a pronounced negative impact on the appraised life remaining of the home. A meter of sinking causes a reduction of 10.7% in the appraised remaining life, and the average home loses 3% of its remaining years of life every 10 years at average sinking intensity.

Our survey corroborates this with correlational evidence that sinking is associated with damaged structures. Columns (2) through (4) regress indicators for whether a respondent reports different structural issues in their home on the sinking on a respondent's unit over the last five years.[41] Column (2) shows that a meter of sinking on the respondent's home over the last 5 years is associated with an increase in the probability of reporting severe structural issues by 44% relative to the mean in the low-sinking strata and increases the probability of reporting issues with cracks in walls by 81%.

Do residents invest in maintenance in response to these damages? Table 3 reports the correlation of survey measures of maintenance investments with sinking on the respondent's plot. While sinking is not associated with extensive margin increases on the probability of incurring maintenance expenses, we do find that a meter of sinking in the last five years is correlated with a large increase of 60 log points in spending on maintenance. Consistent with this, realized sinking is also associated with an a 55 log point fall in the respondent's

[41] We choose this measure because it is observable for all respondents, as survey participation conditioned on having resided at least 5 years in the location. While property fixed effects difference out the time-censored sinking before 2007 in our appraisal data, this is not possible in the cross-sectional survey results.

assessment of when they will have to make their next "large investment" in the home, suggesting that residents are pulling forward regular maintenance in response to sinking.

Finally, the last column of Table 3 tests whether subsidence induces complementary investments that are not related to subsidence (additions, remodeling, electric work, painting, or waterproofing), as these could bias our estimates towards zero if they increase the market value of the home. We find no relationship between sinking and complementary investments, assuaging concerns that these cause us to underestimate the direct costs of subsidence.

5.1.5 Mechanisms: Uneven sinking across neighborhoods

So far we have focused on the impact of plot-level sinking on home values, but uneven sinking across neighborhoods also has the potential to reduce the desirability of a property through its impact on public infrastructure. We study this by introducing our measure of the unevenness of sinking over space into our main specifications.

Table 4 reports results. Column (1) reports the result of estimating the repeat appraisal design on plot-level sinking and our measures of uneven sinking. Uneven sinking across cells has a negative impact on prices, with the average change of 0.069 meters in relative altitude across grid cells from 2007 to the year of sale causing a 1% fall in appraised values. Columns (2) and (3) use survey reports of infrastructure issues in the neighborhood to estimate the correlation of uneven sinking since 2007 with the probability that a neighborhood experiences flooding or fissures in pavement or roads. Column (2) shows that a respondent is 1.8% more likely to report that their neighborhood has issues with flooding at the average relative altitude change of 0.092 meters, though unevenness across tracts does not predict road fissures (Column (3)).[42]

[42]Note that plot-level sinking has no significant relationship with the probability of these infrastructure issues, a finding consistent with intuition from civil engineering studies (Hu et al., 2013; Secretaría de Gestión Integral de Riesgos y Protección Civil, 2018) that the damages caused by subsidence on macro-infrastructure are driven by variation in sinking over space.

5.2 Is future sinking capitalized into home values? Evidence on θ

Within-property variation in expected future sinking is not priced by homebuyers, but perhaps buyers use coarser information to infer future risk. For example, if buyer expectations are better approximated by a property-level static sinking rate, then property fixed effects and realized sinking would both absorb this information. We test for this by relaxing the property fixed effects used in Equation (14), and replacing them with controls for observable hedonic characteristics and coarser 1-kilometer grid cell fixed effects. We limit estimation new builds which have no realized sinking. Table 5 reports results. We find some suggestive evidence of partial capitalization, though results are imprecise and no specification produces statistically significant results. However, comparing magnitudes to the marginal value of a meter of realized sinking (-0.046) would imply a value of θ ranging from 0.3 to 0.80.

Our survey offers additional direct evidence on how homebuyers internalize expected future sinking to a home. Survey respondents are generally familiar with the issue of subsidence; before mentioning the word "sinking" or "subsidence" in our survey, 55% of respondents listed subsidence among their answers when asked to name the likely causes of the deteriorated state of a dilapidated building from a generic photograph.[43] Despite this general awareness, survey respondents report using little *ex-ante* available information to infer risk of future sinking to their home. Figure 5 plots the share of different responses to the question, "How can you know if a home will have issues with sinking in the future?", separated by respondents in high and low risk boroughs.[44] 21% of respondents say it is impossible to know in advance whether a home will have issues with sinking, and despite the strong spatial correlation in sinking, only 20% report that you can infer risk by knowing the home is in a risky area. Zooming out to any ex-ante available information, such as consulting

[43] Additionally, only 18.5% of respondents could not provide a guess when asked where in the city has the most issues with subsidence.

[44] This question was free response, and we prompted respondents for more answers until they indicated they could not come up with additional responses.

an expert or watching the news, this rises to only 42% of respondents overall.

Not only are residents inattentive to *ex-ante* available information about sinking on average, but they show significant sorting on this inattention and surprise about its impacts. We observe in Figure 5 that residents of high-sinking areas are more likely to report future sinking is unknowable and are less likely to use geography to infer risk, a pattern consistent with the more informed using information on risk sort into safer areas. Figure 6 provides additional evidence of sorting on information. Sinking intensity is strongly negatively correlated with the probability of using *ex-ante* available information to infer sinking risk, and positively correlated with the probability a respondent reports they have been surprised by the severity of sinking on their plot.[45]

Consistent with sorting on information, we find that capitalization of future risk is stronger in boroughs with *less* sinking in our appraisal data. Figure 7, Panel (a) plots the capitalization terms $\widehat{\gamma\theta}$ estimated by decile of average sinking rate at the 1-kilometer grid level using the new builds specification from Table 5. While low-sinking boroughs actually have capitalization rates in line with rational expectations ($\widehat{\gamma\theta} \approx \widehat{\gamma}$), it is exactly the boroughs that sink the most on average that do not price this future sinking. By contrast, Panel (b) shows that realized sinking is priced fairly evenly across the sinking intensity of the area, suggesting that this pattern does not reflect heterogeneity in preferences over sinking, but rather how well future sinking is internalized.

Finally, residents also underestimate how much sinking they have experienced *ex-post*. Figure 8 plots respondents' belief about how much their property has sunk over the last five years against our satellite measure, and we find that the slope of this relationship is only 0.05. This highlights that even in retrospect respondents have little idea about how much their property has sunk in quantitative measures, and further highlights the difficulty

[45] While sorting on education or socioeconomic status may explain some of this effect, it does not explain the full relationship, and our results are robust to controlling for respondent and housing unit characteristics, including the education of the respondent (Supplementary Table A4). Furthermore, Supplementary Table A5 confirms that these effects are similar (though noisier due to the small sample size) when limiting to residents who moved within the last 10 years, assuaging concerns that results are driven by selective recall or habituation.

of accurately perceiving slow-moving hazards that are not experienced as dramatic events.

Overall our survey findings suggest that respondents face substantial information frictions about subsidence, and that these frictions are particularly related to the ability to infer the probability of future subsidence absent realized sinking. We also find evidence that residents sort on information about subsidence, with less informed residents being more likely to live in high-sinking properties. In Appendix A3 we explore other potential determinants of under-capitalization, finding no evidence that factors such as risk preferences or discount rates explain our findings, and note the absence of subsidized insurance options or large-scale bailout programs in this context that could explain inattention to sinking. These patterns echo insights from papers studying under-capitalization of hazards in the United States, and suggest that policies such as mandated disclosure could potentially mitigate the pervasive pricing frictions we estimate.

5.3 Supply response to subsidence

The supply model highlights the role that realized subsidence plays in increasing the probability of development by lowering the opportunity cost of redeveloping existing, degraded units. In this section, we exploit this relationship to estimate the supply elasticity using the responsiveness of re-development to realized sinking and prices.

5.3.1 Reduced-form evidence for model predictions

We begin with reduced form tests of our model prediction that realized sinking draws in development. First, we use our matched development-appraisal data to test whether homes that have sunk more are more likely to be sold for redevelopment. Table 6 reports the results of regressing of an indicator for an appraisal being redeveloped on the sinking it has experienced to date and a battery of property characteristics. We include grid cell by year by 5-year age bin fixed effects, effectively comparing homes of similar age sold in the same market (grid cell x year), but with different realized sinking. Strikingly, realized sinking

strongly predicts whether an appraisal is for the purpose of re-development, with a meter of sinking increasing this probability by 1.7 percentage points over a mean probability of 0.45%. This pattern is also observed if we use pixel-level development instead of the matched-appraisals, underscoring that this finding is not a result of errors in matching appraisals to new developments (Supplementary Table A6).

5.3.2 Estimating the supply elasticity

We estimate the supply elasticity using the model-derived redevelopment decision given by Equation (8), which tracks the response of development to subsidence. This decision depends on plot profits π^*_{jt}, which are a function of two different prices: the price of existing, sunk units p_{jt} and the price that new units that have not sunk *yet* fetch, $p_{jt}(0)$. These prices are not simultaneously observable, so we transform this expression into a function of a single price using the decomposition given by the demand structure to separate realized subsidence from the rest of the price: $\log(p_{jt}(0)) = \log(p_{jt}) - \gamma S_{jt}$. Doing so gives:

$$d_{jt} = 1\left(\log(\tilde{\eta}) + \eta \log(p_{jt}) - \gamma(1+\eta)S_{jt} - \log(H_{jt-1}) - \log(D) + \tilde{\xi}_{jt} > 0\right) \tag{16}$$

where $\tilde{\xi}_{jt} = -\eta \log(\xi^S_{jt}) \sim \mathcal{N}(\mu, \sigma^2)$.

Identification is challenged by the endogeneity of cost shocks $\tilde{\xi}_{jt}$ with prices. Supply shocks cause movement along the demand curve in equilibrium, which biases elasticity estimates that ignore this effect toward zero. Our solution is a control function approach. Following Petrin and Train (2010), we impose additional structure on the distribution of cost shocks and identify a demand shifter as an instrument for prices, which allows us to derive a control function that absorbs the endogenous part of the error term.[46] We propose the

[46]Control functions are numerically equivalent to IV estimation in the linear case, and can be extended to non-linear estimation under certain conditions. Appendix A4.5 provides a proof of the validity of using a control function in this context (Blundell and Matzkin, 2014).

following first stage equation derived from our demand model:

$$\log(p_{jt}) = \Pi \text{Rel. Altitude Change}_{jt} + \gamma S_{jt} + \alpha_{n(j)} + \alpha_{z(j)t} + X_{jt}\beta_1 + \tilde{u}_{jt} \tag{17}$$

where Rel. Altitude Change$_{jt}$ is the relative altitude change of the grid cell containing j from 2007 to t, $\alpha_{n(j)}$ are census tract $n(j)$ fixed effects, $\alpha_{z(j)t}$ are year by borough fixed effects, and X_{jt} are our hedonic controls from previous specifications, plus 5-year age bin fixed effects.[47] We assume that $\tilde{\xi}_{jt}$ is distributed joint normally with log prices, with constant variance σ and marginal expectation $\mathbb{E}(\tilde{\xi}_{jt}|\tilde{u}_{jt}) = CF(\tilde{u}_{jt};\phi)$ for some parameter ϕ. Then we can re-write our redevelopment condition in Equation (16) by adding and subtracting the control function and adding the controls from the first stage:

$$\begin{aligned} d_{jt} &= 1\left(\frac{1}{\sigma^*}\tilde{V}^S_{jt} > 0\right) \\ \frac{1}{\sigma^*}\tilde{V}^S_{jt} &= \frac{\eta}{\sigma^*}\log(p_{jt}) - \frac{\gamma(\eta+1)}{\sigma^*}S_{jt} + \frac{1}{\sigma^*}CF(\tilde{u}_{jt};\phi) + \tilde{\alpha}_{n(j)} + \tilde{\alpha}_t + X_{jt}\beta_2 + \frac{\tilde{\xi}^*_{jt} - \mu^*}{\sigma^*} \end{aligned} \tag{18}$$

Our distributional assumptions imply that $\tilde{\xi}^*_{jt} = \tilde{\xi}_{jt} - \mathbb{E}(\tilde{\xi}_{jt}|\tilde{u}_{jt})$ is normally distributed with mean μ^* and standard deviation σ^*. Thus this is estimated via Probit, including the control function as a regressor.[48] Because of the assumption of joint normality, the control function is a linear function of the first stage residuals, i.e. $CF(\tilde{u}_{jt};\phi) = \phi\widehat{\tilde{u}_{jt}}$, and we show robustness to mis-specification by showing that our results are unchanged when including higher order polynomial approximations of this function.

To be considered a valid demand shifter, uneven grid sinking must satisfy the exclusion restriction: it must influence redevelopment solely through its impact on prices. This means the sinking cannot have a direct effect on development costs conditional on controls. The

[47] We include census tract fixed effects to maintain consistent standard error clustering in bootstrap resampling, but results are similar when using grid cell fixed effects. All estimates exclude fixed effect cells with zero variance in the outcome to maintain consistent sample sizes across linear and non-linear models.

[48] See Appendix A4.5 outlines a proof that the new error term $\frac{\tilde{\xi}^*_{jt} - \mu^*}{\sigma^*}$ is mean independent of prices given a valid instrument.

inclusion of census tract fixed effects controls for unobserved, location-specific supply factors, such as local construction code requirements which vary by seismic zone. These plus the control for realized sinking together help address the concern that soil characteristics may be correlated with relative altitude change. We provide empirical evidence in Section 5.4.2 that trends in uneven sinking are uncorrelated with other shocks that influence supply.

Table 7 reports the results of estimating Equation (18). Confidence intervals are estimated via clustered bootstrap to account for the fact that the control function is estimated in the first stage. Column (1) reports the first stage results from regressing log prices on uneven grid cell sinking and controls. Uneven sinking lowers prices conditional on tract fixed effects and unit-level controls, and with an F-statistic of 19.77 the first stage is sufficiently powered.

In line with model predictions, all estimates find that redevelopment responds positively to prices and realized sinking. Interpreting the linear IV estimator in Column (2), each meter of realized sinking on a plot increases the probability of re-development by 3.9 percentage points- a large increase over the mean probability of an appraisal getting redeveloped in this sample of 2.7%. Results are qualitatively similar when looking at the average marginal effects from the probit model in Column (4). This reflects the strong depreciation effect that realized sinking has on the structure, and our finding that sinking has a large negative impact on the useful life remaining of buildings.

Columns (3) and (4) implement model-consistent estimation via control function. While the magnitude of the coefficients is not interpretable due to the unidentified scale parameter, Equation (16) implies that we can estimate the elasticity as $\eta = \widehat{\varepsilon}/(1-\widehat{\varepsilon})$, where $\widehat{\varepsilon} = -\beta^p/\beta_S \times \gamma$.[49] Doing so implies a supply elasticity of 0.28, and this estimate is robust to including a more flexible polynomial approximation for the control function in Column (4). We see that the control function is crucial for addressing bias by comparing these results to

[49] Confidence intervals are calculated overestimates of η in each bootstrap sample, and in each sample γ_b is drawn from a normal distribution centered around the point estimate with standard deviation equal to the standard error of this estimate.

estimates that exclude this control in Column (5), which implies an elasticity near zero. Our estimate of η =0.28 is in line with recent papers that measure within-city housing supply elasticities for the United States (Baum-Snow and Han, 2024; Rollet, 2025).

5.4 Validation and robustness

5.4.1 The direct cost of subsidence

Table A7 reports robustness of our main estimates to different ways of conducting inference and addressing potential sources of bias.

Inference Column (1) of Table A7 reproduces our preferred specification, but estimates standard errors using a 2-kilometer radius for calculating Conley standard errors. Precision is similar to using census tract fixed effects, reflecting the low spatial correlation of residualized sinking. Results are also robust to modeling potential measurement error in subsidence; Figure A5 plots the distribution of estimates and p-values from a Monte Carlo simulation that models noise in the calculation of the subsidence rate, showing that accounting for this uncertainty does not change results.[50]

Potential confounders Column (2) of Table A7 shows that our findings are robust to the inclusion of additional controls for time trends based on the type of structure, number of bedrooms, and binned area of construction, assuaging concerns that the valuation of certain attributes were evolving over time in a way that was correlated with sinking trajectories.

Column (3) tests whether our results are driven by changes in perceptions of seismic risk, which is very correlated with sinking in the cross-section. We do this by estimating Equation (15) excluding the period after September 2017, when a major earthquake hit Mexico City. If anything the effects are stronger when limiting to the pre-2017 period, underscoring that our estimates pick up changes in sinking, not changes in perceived seismic risk.

In Column (4) we verify that results are not driven by mechanical appraiser valuation of

[50]We assume errors are distributed normal with mean zero and standard deviation of 10 millimeters, a conservative estimate of the measurement error for SAR Interferometry in other settings (Liu et al., 2020; Bawden et al., 2023).

subsidence by dropping localities above the median in terms of the locality's 75th percentile of differences between appraised and transaction values in the R04 data, and find that results are robust to this restriction.[51]

We conduct a placebo test that tests whether the subsidence of a plot's neighboring pixels affects prices. This is informative about whether general neighborhood sinking is correlated with evolutions in home values, and also tests whether the increased probability that the plot's neighbors are re-developed leads to price effects through gentrification. Column (5) shows that sinking on neighboring plots is not correlated with housing values, giving confidence that we are measuring the impact of damage to the home itself, not other trends in the neighborhood.

We rule out that other economic shocks that may affect home values are correlated sinking using the confidential version of the Economic Census in Mexico, which reports comprehensive information on all fixed-establishment firms (formal and informal) every six years. While we do not know the time a firm has been in a particular structure, we can test whether sinking intensity (the sinking rate) is correlated with changes in firms' employment, sales, or sectoral composition. Table A8 reports results; the sinking rate is not correlated with firm sales, employment, or probability of being in a particular sector conditional on census tract and borough by year fixed effects, suggesting that other economic trends are not correlated with the micro-geography of sinking.

Mortgage penalties Finally, we provide evidence that changes in appraised values are not driven by mortgage penalties on sinking homes. If banks charge higher interest rates on homes that are prone to sinking, then the changes in values we estimate may represent discounts that the market offers to offset these rates. Table A9 in the Appendix estimates the cross-sectional correlation of the sinking intensity of the census tract with the interest rate charged on mortgages on that tract using the R04 data, finding no evidence that banks

[51]The median 75th percentile is 0.08%. We use the 75th percentile because the median difference in prices at is zero for all localities.

assign differential interest rates to high sinking areas.[52]

5.4.2 Supply estimation

Identification in our supply estimation requires that redevelopment trends be similar across areas, both in terms of plots experiencing high versus low sinking and in neighborhoods with uneven versus uniform sinking. Figure A3 showed that the amenities of new units were not systematically evolving in a way correlated with expected sinking, but we also test this assumption by comparing the development trends of pixels that were never sinking quickly with those that subsequently started sinking quickly later in the panel. To formalize this, we classify a pixel as "high sinking" using a cut-off based on the 2020 distribution of sinking rates and define the pixel as "treated" starting in the first year its sinking rate reaches that threshold. Figure A6 reports Sun and Abraham (2021) event-study estimates, which exclude pixels that were always fast- or unevenly-sinking. The results confirm that the not-yet-sinking pixels were trending similarly to the never-sinking pixels before the sinking accelerated.

6 Counterfactuals and policy implications

We use the parameters estimated in previous steps and direct calibration of the remaining parameters to estimate the total costs of sinking and the role pricing frictions play in driving these costs. The full set of estimated and calibrated parameters is summarized in Supplementary Table A10.

[52] A limitation of this test is that we cannot observe mortgages that were not approved by the bank, so there may be extensive margin mechanisms through which the mortgage market addresses subsidence that we cannot identify.

6.1 The total cost of sinking, benefits of alleviating frictions

We estimate the costs of subsidence and the benefits of alleviating information frictions by calculating the difference in welfare between the observed equilibrium and a counterfactual equilibrium according to Equation (13). Appendix A4.6 provides additional details. We estimate the total costs of subsidence for different levels of capitalization by comparing welfare in the observed equilibrium to one in which all expected future sinking is set to zero ($\mathbb{E}_t(s_{j\tau}) = 0 \, \forall j, \tau > t$). By varying θ, we can examine how pricing frictions exacerbate the total costs of subsidence by over-allocating housing into risk areas.

Figure 9 depicts the gains from these counterfactuals, expressed in billions of 2018 USD. Panel (a) shows how results vary as θ goes from zero to one. At no capitalization, homeowners reap all the benefits from stopping subsidence, as they benefit from paying efficient prices for this depreciating asset. As $\theta \to 1$, homeowners no longer benefit from sinking stopping, as they make zero profits on their real estate investments. Instead, renters and the developer benefit, as stopping sinking results in hazardous areas of the city becoming more attractive, which leads to more units built, and higher quality units as they do not sink. At a $\theta = 0.5$, the share of survey respondents that use any ex-ante available information to predict sinking, subsidence costs Mexico City 8.27 billion US dollars, representing about 4.4% of all housing value in the city.

Panel (b) decomposes the welfare gains and losses among market participants in a counterfactual that simulates alleviating pricing frictions by comparing welfare under $\theta = 1$ relative to $\theta = 0.5$. Eliminating these frictions generates a gain of \$3.96 billion USD, or almost half of the gains from completely eliminating subsidence. With no pricing frictions, homeowners again gain as they go from making negative to zero profits. Housing developers lose out in the no-frictions counterfactual, as they benefit from over-valuations under frictions. Renters dislike the higher rents they pay when housing supply falls, as the amenity value they experience is unchanged if sinking continues. This disparity in winners and losers from alleviating pricing frictions contrasts with the benefits of stopping sinking, which

accrue to all market participants though to different degrees.

The extent to which pricing frictions distort the geography of housing can be seen in Figure 10, which maps changes in housing density and prices under the no pricing friction counterfactual. Alleviating pricing frictions lowers housing density by up to 2.5% in the highest-risk areas, and these areas see a sharp fall in equilibrium prices. This underscores how pricing frictions aggravate the total costs of environmental hazards by allocating too much housing capital into risky areas.

6.2 Using welfare estimates to value alternative policies

The previous model counterfactuals provide an estimate of the total cost of subsidence. In this section we use this cost estimate in back-of-the-envelope exercises that allow us to value alternative policies that address groundwater depletion.

6.2.1 The external cost of groundwater pumping

In order to price groundwater pumping, the externality-generating action, we must convert total costs to costs per unit of water pumped. We do this using calibrated hydro-physical relationships between aquifer storage and subsidence in the Mexico City Valley; Appendix A5 provides details. Combining our total cost estimate with the elevation-to-pumped-volume conversion implies that subsidence costs the city $0.22 USD per cubic meter pumped in annual losses. Adding up the dynamic benefits of leaving water in the aquifer implies a total present value of $3.90 per cubic meter.

Most groundwater pumping in Mexico City is for residential use; how much more would residents pay for water if these external costs were passed onto water bills? Comparing this cost to water tariffs in Mexico City, we estimate that the external cost of groundwater pumping amounts to a 173-968% increase over the average marginal rate paid by residents, and would add 104-948% to the average total bill (Supplementary Table A11).[53] These are

[53]We assume that 60% of the water bill is taxed, the average share of water in Mexico City that comes

large increases, and would only alleviate the externality if water demand responds elastically to price.

6.2.2 Cost-benefit analysis of other recharge policies

Our cost estimates can also be used to value interventions that would directly mitigate subsidence by facilitating aquifer recharge. These interventions fall into three categories: Directed recharge of the aquifer via injection of wastewater; Investment in alternative water sources (surface water or recycled water) that would substitute groundwater pumping; and investments in repairing leaks in the existing infrastructure which would lower total water demanded. Each of the interventions we study have been implemented or proposed at some point in Mexico City as a way to offset groundwater decline.[54]

We find substantial promise for groundwater mitigation policies to generate net economic benefits. Figure 11 depicts the results of these analyses, plotting the benefit to cost ratio of each intervention. Implementing all interventions with a benefit to cost ratio above one would imply offsetting 38% of the city's annual groundwater pumping. Repairing leaks saves the least amount of water, but its benefits dramatically outweigh the costs assuming that these repairs do not take away from recharge. While some alternative water sources show high returns on investment (Taxhimay, Oriental-Libres), the costs of others do not appear to exceed benefits. Wastewater injection also has a high benefit to cost ratio, and this is likely a lower bound as the benefits of injecting wastewater will be larger if spatial targeting is taken into account.[55]

from groundwater extraction.

[54]These analyses should be taken as suggestive; while policy interest and previous investments in each of these options over the last 30 years in Mexico City provide some idea of potential costs (SACMEX, 2012), current and accurate cost data for these types of investments would improve these estimates. Detailed documentation of the assumptions made and cost sources for each of these are available in Appendix A5.2.

[55]We assume that wells evenly affect subsidence across the city in these calculations, but if wells are built near areas that are subsiding quickly then the imperfect connectivity of the aquifer means that this would have a larger local impact on subsidence, potentially increasing the gains even further.

7 Conclusion

We conduct a comprehensive valuation of the costs of subsidence in Mexico City, finding that subsidence costs Mexico City over $8.27 billion USD. Despite the large costs that realized sinking imposes on homes, predictable future sink is unpriced in the market. We estimate that that pricing frictions drive almost half the total costs of sinking, as they cause an over-supply of housing in hazardous areas.

Importantly, our estimates represent a lower bound on the costs of subsidence as they exclude city-wide costs that are shared among residents. For instance, our estimates do not capture expenditures incurred by the city government to repair water main breaks or sinkholes or replace macro-infrastructure. Even so, our estimate likely captures the largest part of the costs of subsidence, as the costs to housing we estimate are orders of magnitude larger than, for example, the cost of the Cutzamala surface water system ($0.58B USD in present value), and the cost of the recent drainage system replacement ($65 million USD).

More generally, our framework provides a systematic way of thinking about re-development in places that experience environmental hazards such as climate impacts, erosion, or floods. Our model highlights the different economic role that *realizations* of environmental hazards play in driving re-development, as compared to expected *future* hazards that drive the aggregate allocation of housing. Our empirical findings show that pricing frictions affecting the valuation of future realizations of environmental hazards, which have been found to be widespread in different contexts, can severely exacerbate the costs of these hazards by allocating too much housing in risky areas.

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Tables and Figures

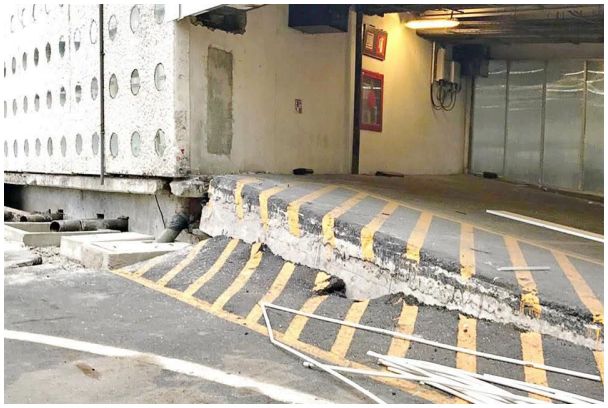
Figure 1: Structures affected by subsidence in Mexico City



(a) Building distorted by sinking, downtown Mexico City



(b) Uneven subsidence on road



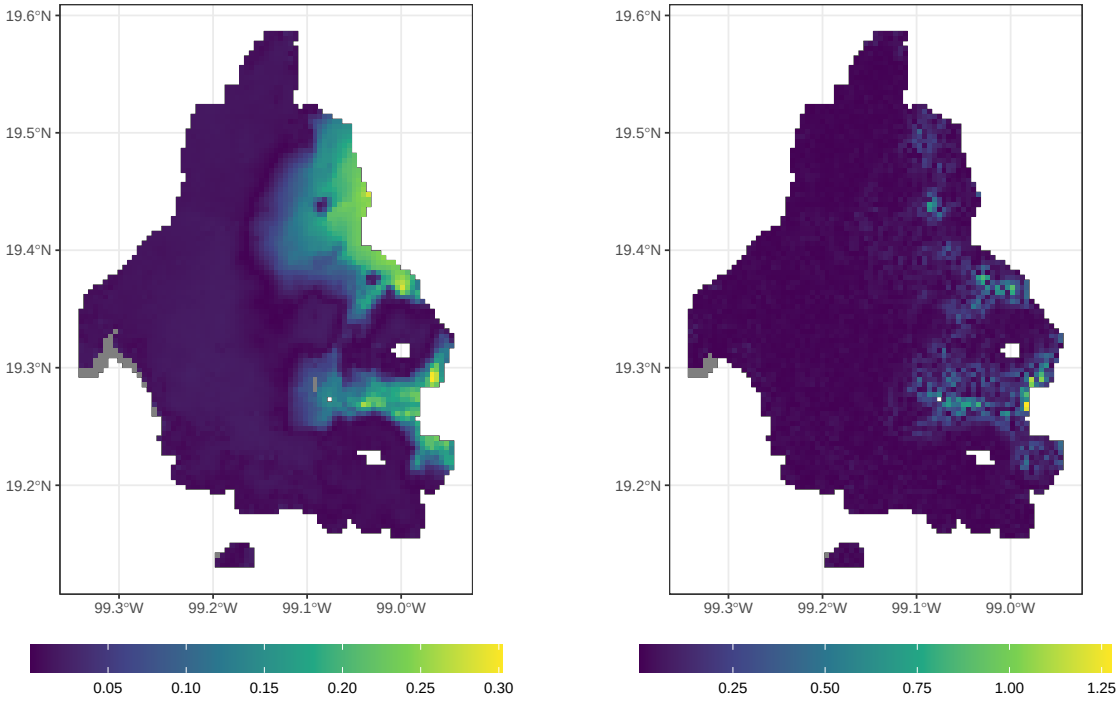
(c) Subsiding building



(d) Tilting wall, Venustiano Carranza

Notes: Subfigure (a) is from the New York Times, article "Mexico City, Parched and Sinking, Faces a Water Crisis". Sub figure (b) from *Cultural Colectiva*. Subfigure (c) borrowed from *Instituto de Ingeniería*, UNAM. Subfigure (d) is from the Mexico City government's Subsidence Atlas (Secretaría de Gestión Integral de Riesgos y Protección Civil, 2018).

Figure 2: Subsidence rates in Mexico City

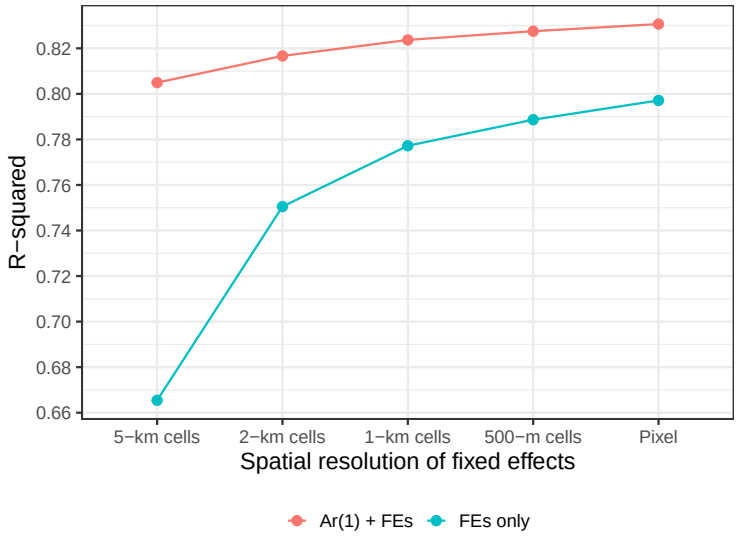


(a) Sinking rate in 2015

(b) Relative altitude change, 2007-2020

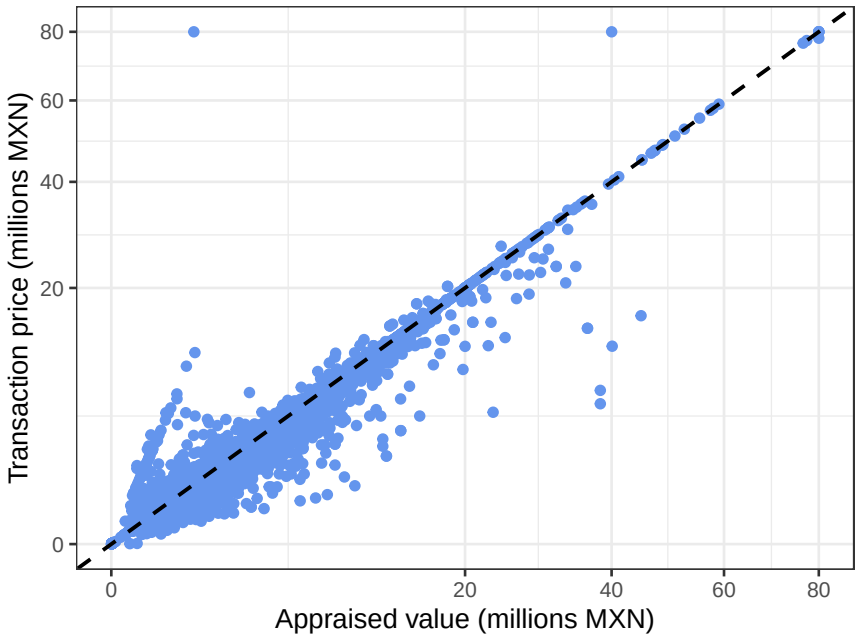
Notes: Map extent limited to grid cells with positive population within the state of Mexico City. Cells in grey do not have data in 2015. Panel (a) plots the sinking rate at the pixel level in 2015. Panel (b) plots the total cumulative relative elevation change from 2007 to 2020.

Figure 3: Predictability of subsidence over space



Notes: Each point represents the unadjusted R^2 of a regression of the pixel-year level sinking rate on a set of increasingly granular spatial fixed effects whose resolution is noted on the X-axis. The regressions generating the points in blue include only these fixed effects; regressions generating the points in pink additionally include the lagged sinking rate as an explanatory variable.

Figure 4: Appraisals and transaction prices



Notes: Each point represents an appraisal-transaction price pair from the mortgage data; a random sample of 50,000 pairs was drawn from the full sample. Differences winzorized at the 99th percentile. The dashed line is the 45° line.

Table 1: Impact of plot sinking on appraisal values

	AR(1)		Perfect foresight	
	3%	6%	3%	6%
Property-level sinking (m)	-0.04618***	-0.04595***	-0.04607**	-0.04574**
	(0.01556)	(0.01558)	(0.01940)	(0.01949)
NPV future sinking	0.01276	0.02170	0.00595	0.00928
	(0.01105)	(0.02116)	(0.09194)	(0.09676)
Observations	129,765	129,765	129,765	129,765
θ	-0.276	-0.472	-0.129	-0.203
SE	(0.258)	(0.491)	(2.029)	(2.169)
ATE: plot sinking	-0.8%	-0.79%	-0.8%	-0.79%
Mean sinking (m)	0.173	0.173	0.173	0.173
Property FEs	X	X	X	X
Borough x Year FEs	X	X	X	X
Vintage x year FEs	X	X	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The outcome in all regressions is the log appraised value of the home. All estimates include property, vintage by year, and borough by year fixed effects. Estimation limited to properties with more than one appraisal. Standard errors clustered at the census tract. The average treatment effect (ATE) is calculated by multiplying the reported coefficient by the average realized sinking in meters. Standard error clustered at the census tract.

Table 2: Sinking and structural issues

	log years left	Structural issues	Cracks: Hallways	Cracks: Walls
	(1)	(2)	(3)	(4)
Property-level sinking (m)	-0.10700**			
	(0.04692)			
NPV future sinking	-0.07356			
	(0.24830)			
Sinking in last 5 years (m)		0.259**	0.114	0.226*
		(0.122)	(0.097)	(0.114)
Observations	129,418	800	800	800
Mean Y	63.311	0.59	0.204	0.28
Data	Appraisals	Survey	Survey	Survey
Repeat transaction controls	X			
Survey respondent controls		X	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Column (1) regresses the log appraised years of life remaining on total sinking on the property, controlling for property, vintage by year, and borough by year fixed effects. In Columns (2) through (5), all outcomes are indicators for reporting the stated issue in their home and the regressor is the total sinking on the property in meters from the year of construction (or 2007, whichever is later) through 2024. Columns (2) through (5) include survey strata, borough, respondent age, gender, and educational attainment fixed effects and are weighted by sampling weights. Standard errors clustered at the census tract.

Table 3: Survey evidence: Sinking and maintenance expenditure

	Any spending (1)	log Spending (000s MXN) (2)	log Years until next big investment (3)	Any spending on non-sinking related issues (4)
Sinking in last 5 years (m)	0.062	0.601**	-0.554**	0.062
	(0.111)	(0.295)	(0.271)	(0.088)
N	625	625	616	623
Mean Y (low sinking)	0.578	17.987	6.707	0.301
Mean X	0.314	0.314	0.308	0.314
Survey respondent controls	X	X	X	X

* p <0.1, ** p <0.05, *** p <0.01

Notes: Outcomes in order of appearance are: An indicator for any spending on maintenance; log of 1+ the total spending on maintenance over the last 6 months in thousands of Mexican pesos; the log of respondent's estimated time in years to the next large investment in the home; and an indicator for the household mentioning any non-sinking category of response: electric work, remodeling, painting interiors or exteriors, additions, water-proofing, or replacing windows. All specifications include survey strata, borough, respondent age, gender, and educational attainment fixed effects. All regressions weighted by sampling weights. Standard errors clustered at the census tract.

Table 4: Neighborhood sinking impacts home values and damages public infrastructure

	log appraised value	Flooding	Fissures
	(1)	(2)	(3)
Property-level sinking (m)	-0.03781**	-0.025	-0.095
	(0.01915)	(0.065)	(0.114)
NPV future sinking	0.00989		
	(0.08826)		
Relative altitude change (m)	-0.15147*	0.124**	0.044
	(0.08963)	(0.061)	(0.161)
Observations	129,765	800	800
Mean abs. rel. altitude (m)	0.069	0.092	0.092
Mean Y		0.647	0.646
Data	Appraisals	Survey	Survey
Repeat transaction controls	X		
Survey respondent controls		X	X

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Column (1) regresses the log appraised value on sinking on the property and the cumulative change in relative altitude, with property, vintage by year, and borough by year fixed effects. Columns (2) and (3) regresses an indicator for the household reporting that their neighborhood has issues with the infrastructure problem in the header on total sinking on the property since it was built and the cumulative relative altitude change from 2007-2024. Columns (2) and (3) include strata, respondent education, age, and gender fixed effects, and regressions are weighted by sample weights. Standard errors clustered at the census tract.

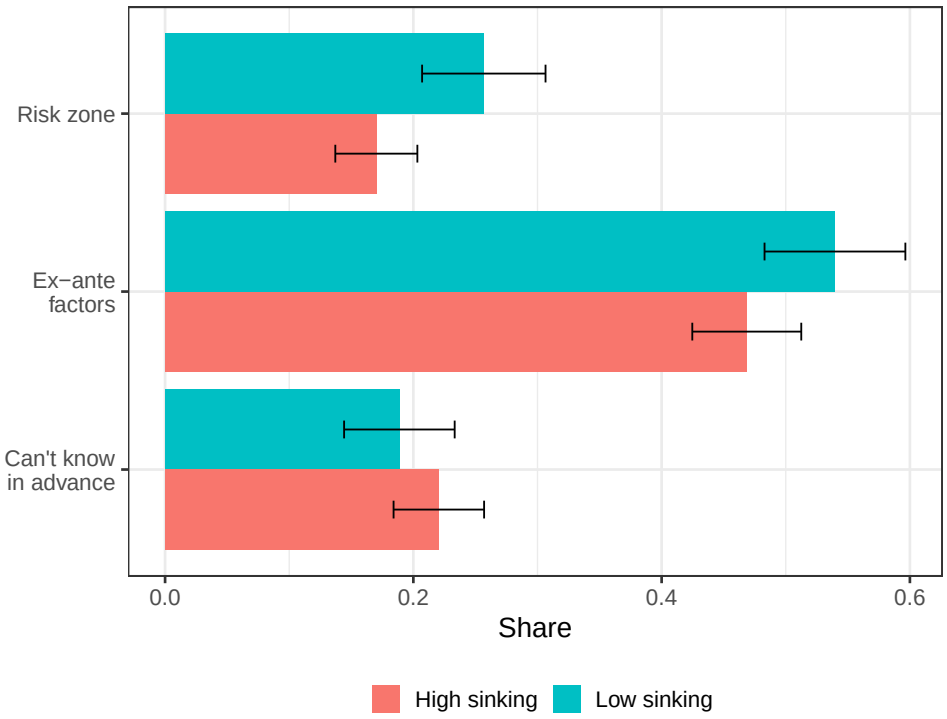
Table 5: Price response to expected future sinking, new builds

	AR(1)		Perfect foresight	
	3%	6%	3%	6%
NPV future sinking	-0.01951	-0.03786	-0.01385	-0.02622
	(0.01416)	(0.02805)	(0.01076)	(0.02086)
Observations	61,444	61,444	61,444	61,444
Grid FEs (1 km)	X	X	X	X
Borough x Year FEs	X	X	X	X
Hedonic controls	X	X	X	X

* p < 0.1, ** p < 0.05, *** p < 0.01

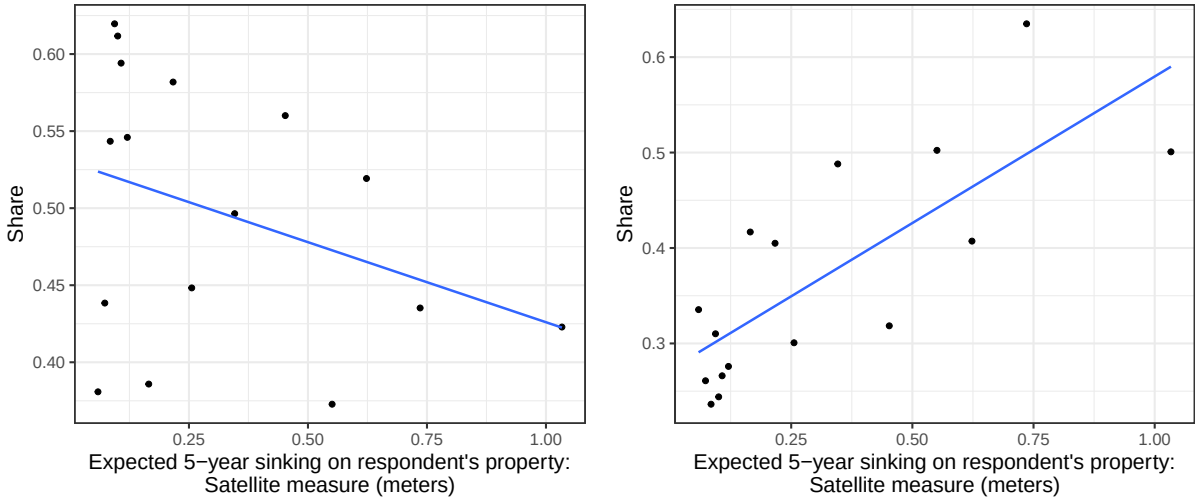
Notes: The outcome in all regressions is the log appraised value. Regressions limited to new builds with more than one appraisal. All estimates include borough by year and 1-km grid cell fixed effects, and control for the area of the plot, the area of construction, and fixed effects for the number of bedrooms, number of bathrooms, number of parking spaces, building class, number of floors, type (single-family, apartment, etc.), and number of elevators. Standard errors clustered at the census tract.

Figure 5: How can you know if a home will have issues with subsidence in the future?



Notes: Each bar plots the average of an indicator for the respondent mentioning a way of knowing that a house will have future issues with sinking in that category, weighted by the population weights and separated by the high-sinking and low-sinking strata. Black bars represent the 95% confidence interval around the mean. "Risk zone" groups any answer that references the location, geology, or risk area of the property. "Any ex-ante" groups any response related to the risk zone, consulting an expert, consuming news, observing the condition of the neighborhood, or investigating the building quality.

Figure 6: Sinking against attention and surprise



(a) Uses ex-ante available information

(b) Surprised by severity of sinking

Notes: Each plot calculates the average outcome within bins of expected 5-year plot-level sinking, estimated as $5\overline{s}_j$ where $\overline{s}_j$ is the average sinking rate on the plot. Number of bins chosen using the Freedman-Diaconis rule. Panel (a) plots the average of an indicator for the respondent reporting they use any ex-ante available information to infer sinking risk (knowing the risk zone, asking an expert/engineer, seeing issues reported in the news, observing issues in the neighborhood, or knowing the construction quality of the home). Panel (b) plots the average of an indicator for the respondent reporting that the issues related to subsidence have been more severe than they anticipated. Linear line of best fit through binned means plotted in blue.

Table 6: Impact of sinking on the probability of re-development

	(1)	(2)	(3)
Sinking on the property (m)	0.0173***	0.0168***	0.0170***
	(0.0045)	(0.0044)	(0.0044)
log Price			0.0017
			(0.0012)
N	221,845	221,833	221,833
Mean Y	0.0045	0.0045	0.0045
Adj. R2	0.423	0.423	0.424
1km Grid x Year x Age bin FEs	X	X	X
Hedonic controls		X	X

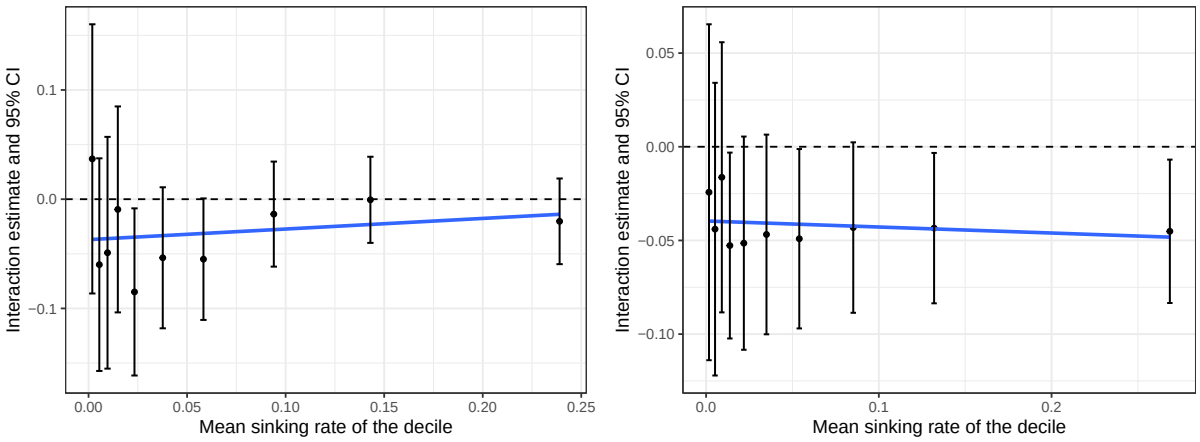
* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The outcome in all regressions is an indicator for the appraisal being associated with a new development. All specifications include year by 1-km grid cell by 5-year age bin fixed effects. Specifications with hedonic controls control for the area of the plot, the area of construction, and fixed effects for the number of bedrooms, number of bathrooms, number of parking spaces, building class, number of floors, type (single-family, apartment, etc.), and number of elevators. Standard errors clustered at the census tract.

Figure 7: Capitalization by sinking intensity

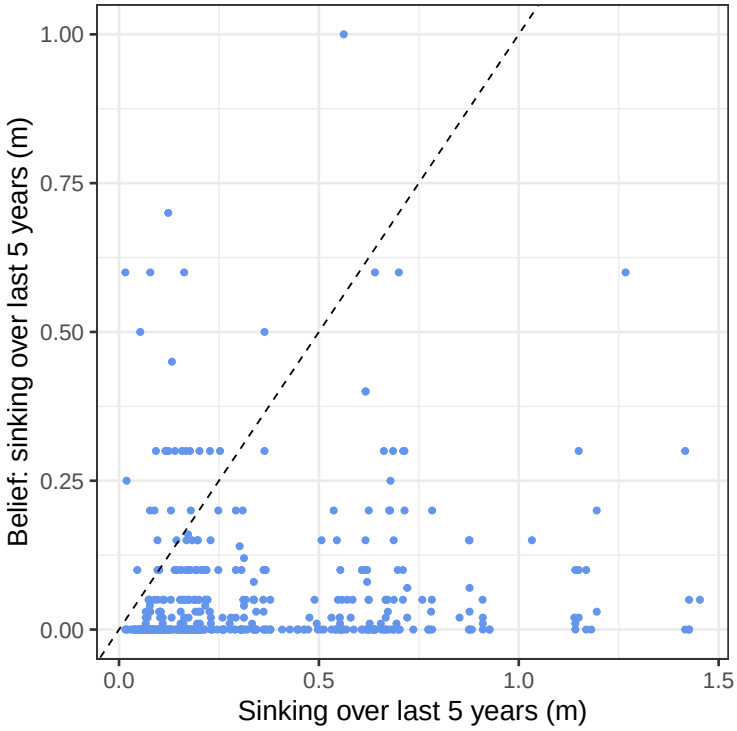
(a) Future sinking

(b) Realized sinking



Notes: Each point with error bar represents the regression coefficient and 95% confidence interval of the interaction of the covariate with a fixed effect for each decile of average sinking at the 1-kilometer grid cell level. The blue line fits a linear regression through the point estimates. Panel (a) interacts fixed effects with the net present value of future sinking specified using an AR(1) process and 6% discount rate, and includes 1-kilometer grid cell and borough by year fixed effects, as well as controls for the area of the plot, the area of construction, and fixed effects for the number of bedrooms, number of bathrooms, number of parking spaces, building class, number of floors, type (single-family, apartment, etc.), and number of elevators. Estimation limited to new builds with repeat sales. Panel (b) interacts fixed effects with realized sinking on the plot, and controls for expected future sinking measured using AR(1) expectations and a 6% discount rate as well as property, vintage by year, and borough by year fixed effects. Estimation limited to properties with repeat sales. Standard errors clustered at the census tract.

Figure 8: Beliefs about past sinking versus satellite measures



Notes: Each point plots the respondent's report of how much their plot has sunk in the last five years against the satellite measure of how much that plot sunk over the same period. The dotted line plots the 45° line.

Table 7: Supply elasticity estimates

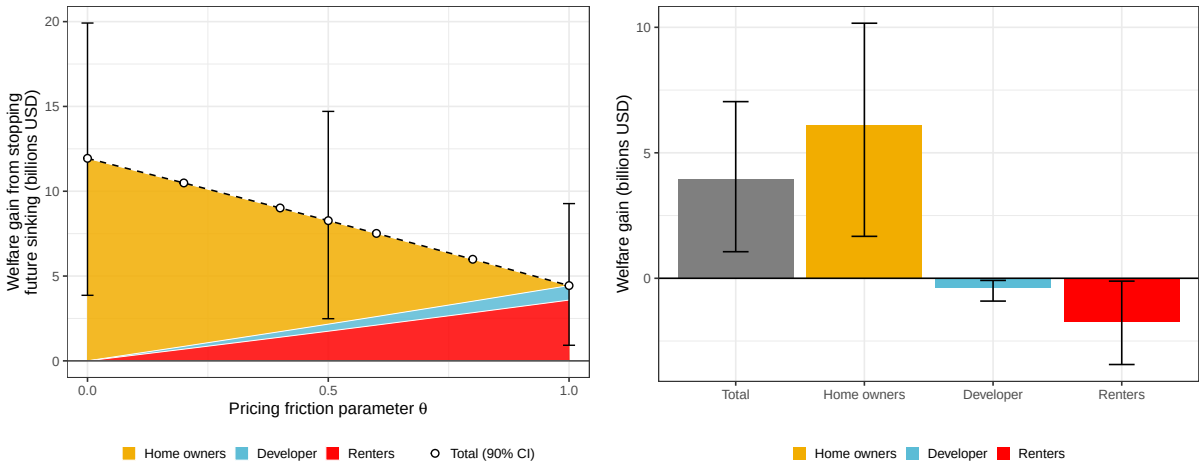
	$\log(p_{jt})$	$\mathbf{1}(\text{Re-developed}_{jt})$			
	First stage	IV	Probit	Probit, Flexible CF	Probit, no CF
	(1)	(2)	(3)	(4)	(5)
Relative altitude change (m)	-0.936***				
	[-1.442, -0.485]				
log Price		0.167	5.935	5.952	0.010
		[-0.258, 0.700]	[1.101, 24.653]	[0.217, 24.719]	[-0.148, 0.215]
Property-level sinking (m)	-0.078***	0.039**	1.654	1.660	0.895
	[-0.127, -0.035]	[0.005, 0.101]	[0.705, 4.867]	[0.720, 4.887]	[0.347, 1.805]
u			-5.927	-6.068	
			[-24.660, 0.016]	[-24.949, -0.304]	
u^2				-0.237	
				[-0.818, 0.122]	
u^3				0.295	
				[-0.046, 0.608]	
Observations	73,367	73,367	73,367	101,888	101,888
F stat (KPR)	16.006				
Estimated elasticity, η		0.167	0.279	0.281	0.001
		[-0.258, 0.700]	[0.005, 0.873]	[0.014, 0.873]	[-0.017, 0.019]

Notes: The outcome in Column (1) is the log transaction price; the outcome in Columns (2)-(5) is an indicator for the plot being redeveloped. Brackets under each estimate report the 95% bootstrap confidence interval. All specifications include borough by year, census tract, and 5-year age bin fixed effects and control for the area of the plot, the area of construction, and fixed effects for the number of bedrooms, number of bathrooms, number of parking spaces, building class, number of floors, type (single-family, apartment, etc.), and number of elevators. Estimation limited to census tracts with variation in the probability of redevelopment. Standard errors clustered at the census tract in the first stage and linear probability model; confidence intervals calculated from bootstrapped sampling of census tracts in Columns (3)-(5), with 1,000 bootstrap samples drawn of 235 census tracts sampled in each.

Figure 9: Welfare benefits of counterfactual policies

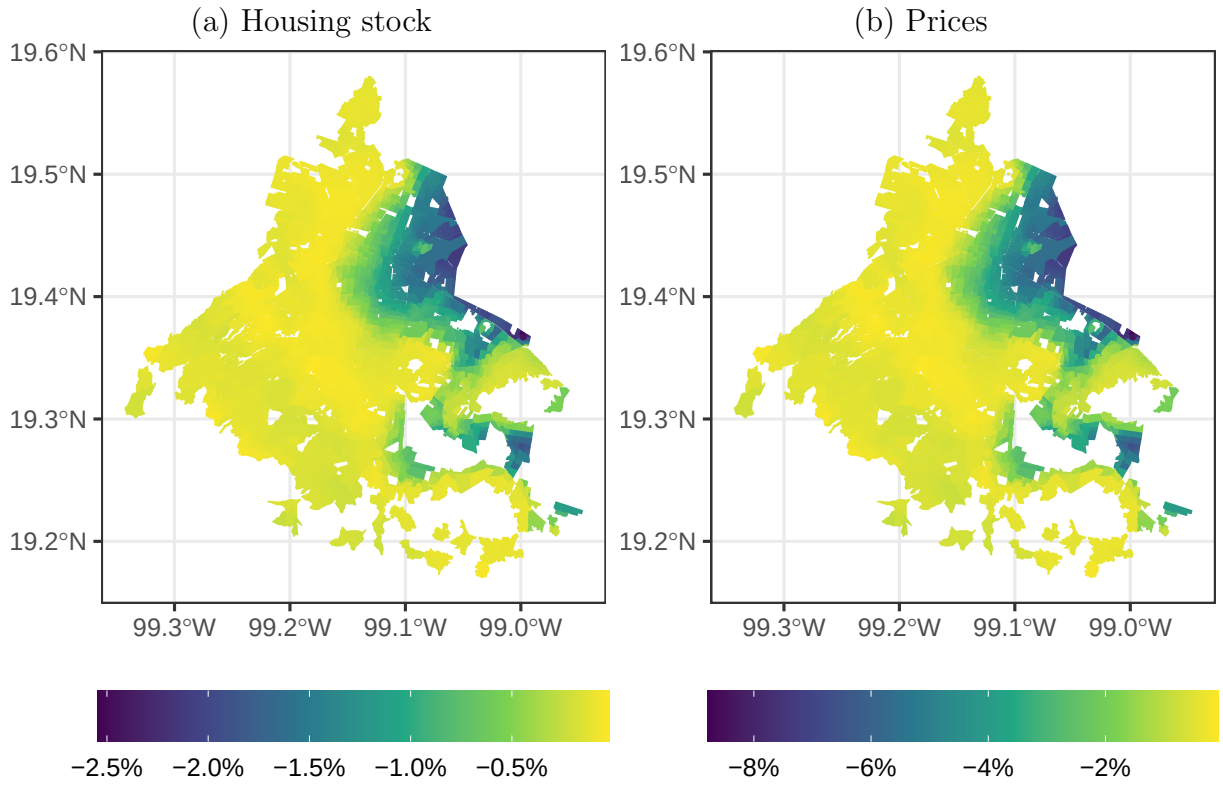
(a) Gains from stopping subsidence

(b) Gains from alleviating pricing frictions



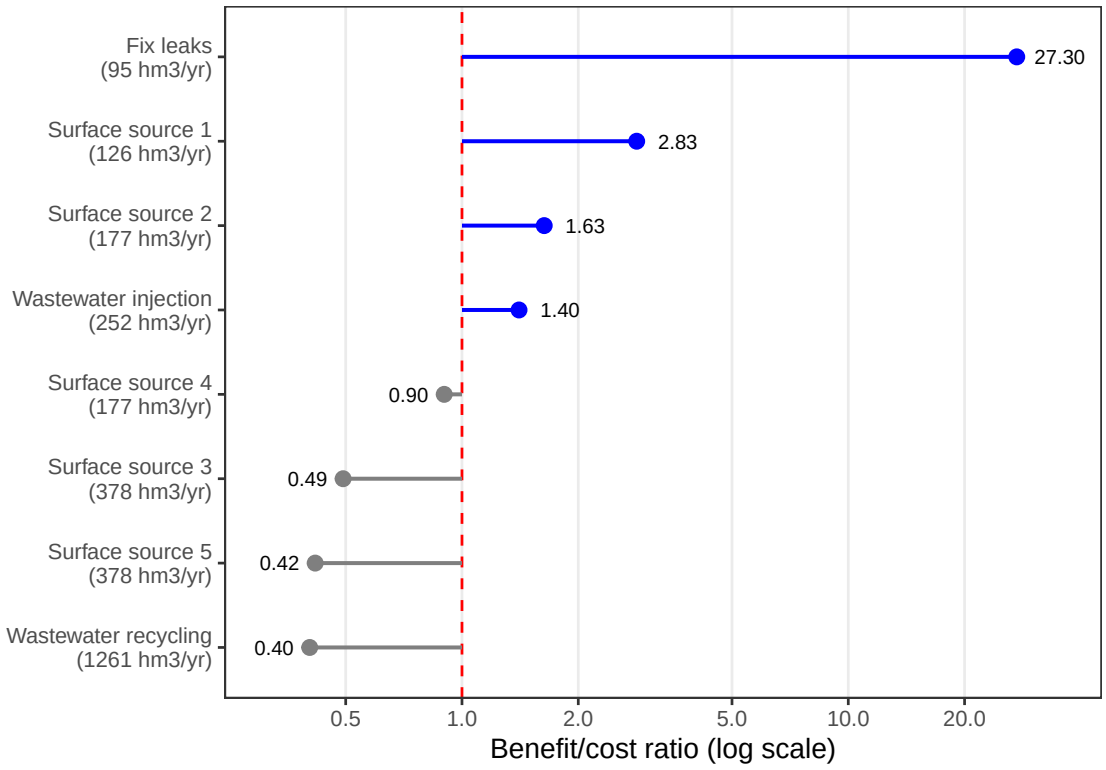
Notes: In Panel (a) the height of each color represents the total welfare gain to that market participant assuming a value of θ given on the horizontal axis, expressed in terms of billions of 2018 USD. "Stopping subsidence" means setting $\mathbb{E}_t(s_{j\tau}) = 0 \forall \tau > t$. Error bars mark the 5th to 95th percentile range of estimates from 500 samples γ and η from normal distributions centered around their point estimates with standard deviation given by the standard error of the estimate. In Panel (b) each bar height represents the change in surplus by market participant for the counterfactual minus a baseline equilibrium, expressed in terms of billions of 2018 US dollars. Error bars mark the 5th to the 95th percentile range of estimates from 500 samples γ and η from normal distributions centered around their point estimates with standard deviation given by the standard error of the estimate.

Figure 10: Spatial reallocation of housing under no pricing frictions



Notes: Fill colors per census tract represent the percentage change in the housing stock (Panel (a)) and equilibrium price (Panel (b)) in the counterfactual calculated with no pricing frictions compared to the observed housing density.

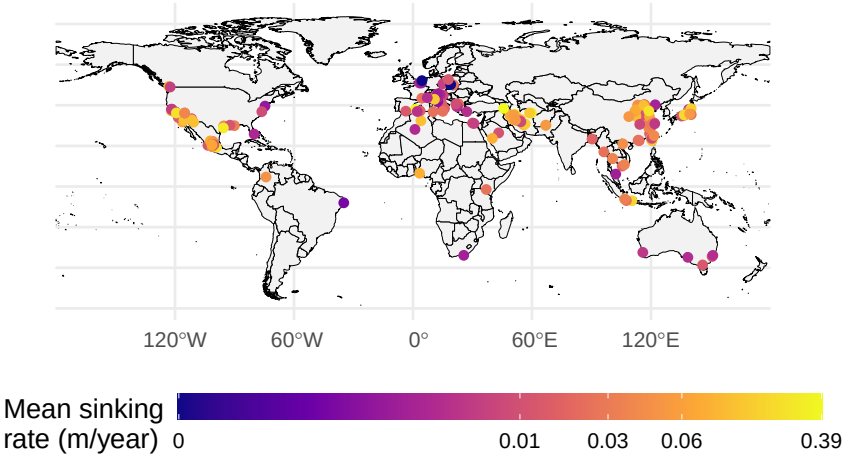
Figure 11: Benefit to cost ratios of groundwater recharge policies



Notes: The potential surface water sources referenced are taken from López-Morales and Mesa-Jurado (2017) and represent the following basins (in order labeled 1-5): Taxhimay, Oriental-Libres, Amacuzac, Temascaltepec, and Tecolutla. 6% discount rate used to calculate present value of costs and benefits. The total benefit of the intervention is in terms of abated subsidence, calculated using a value of $0.22 per cubic meter in annual benefits from abated pumping and bringing the stream of benefits over the life of the intervention to the present value.

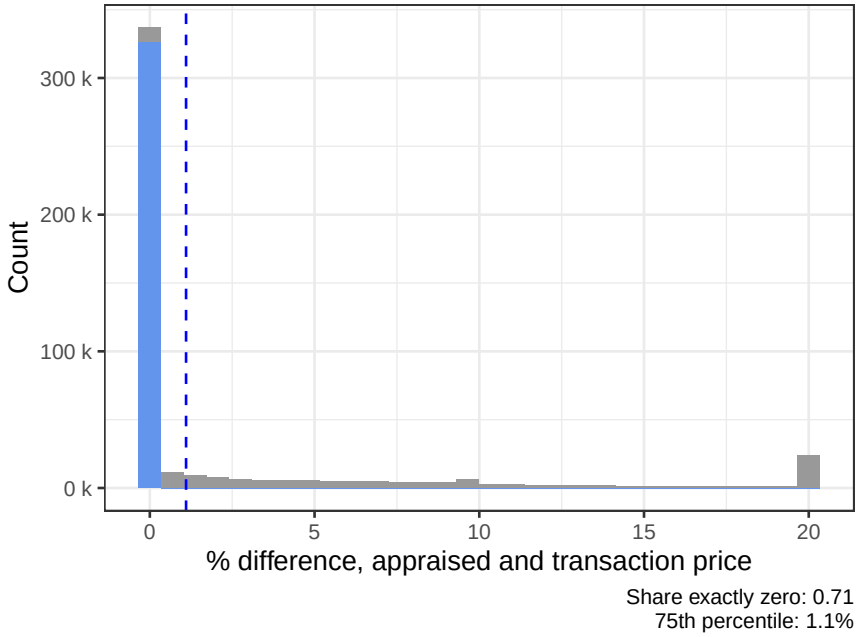
Supplementary tables and figures

Figure A1: Subsidence rates from the geophysics literature



Notes: Figure adapted from data reported in Bagheri-Gavkosh et al. (2021), who survey geophysical studies of land subsidence published before 2021. The median study included in the survey calculated subsidence rates starting in 2001 and ending in 2010. Points are colored by the mean subsidence rate estimated in the study when available, and the maximum rate when that is the only measure reported.

Figure A2: Distribution of deviations of appraised and transaction values

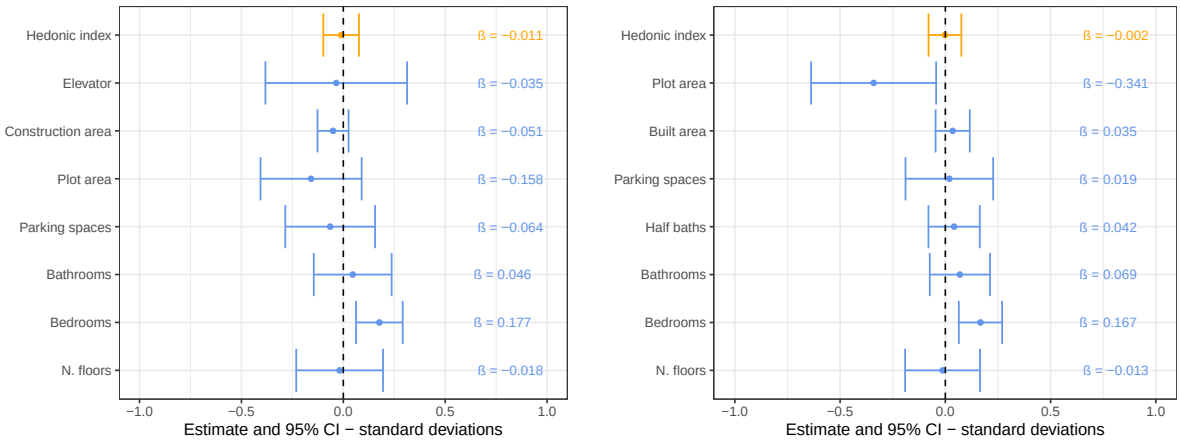


Notes: Height of the bars represents the number of appraisals with the given percentage difference between appraised and transaction values, measured as $100 \times (p^{appr} - p^{transac})/p^{transac}$. Values winzorized at the 99th percentile for visual clarity. Exact zero values colored in blue.

Figure A3: Pre-trends test: Subsidence and market composition

(a) AR(1) expectations

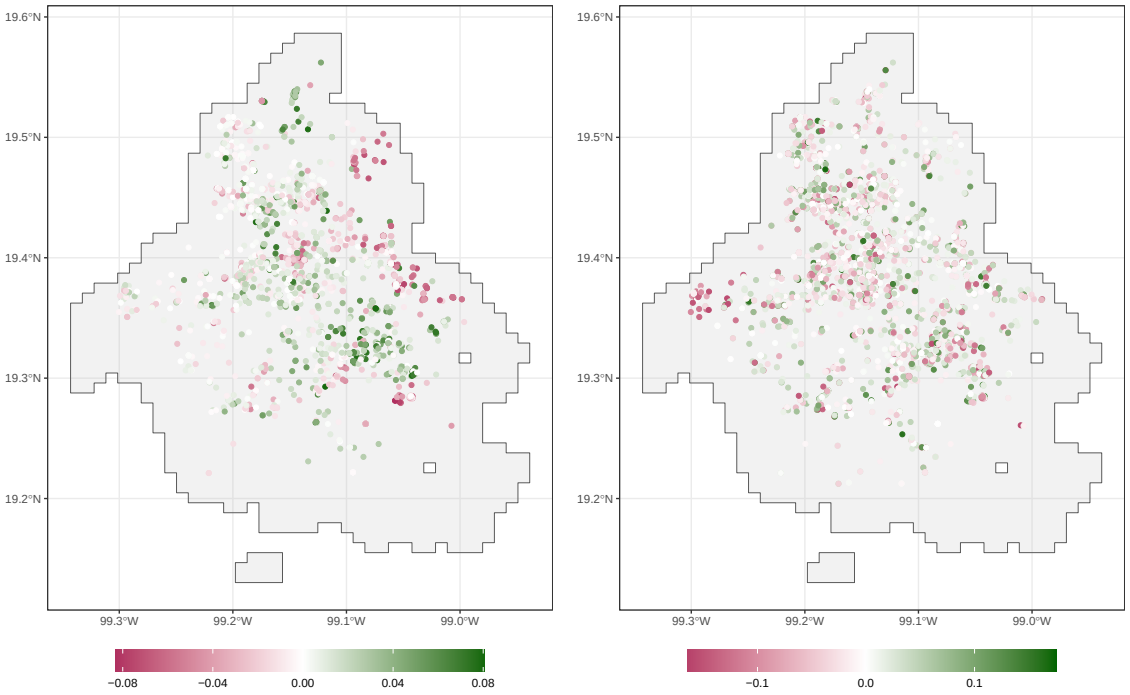
(b) Perfect foresight expectations



Notes: Plots coefficients and 95% confidence intervals for separate regressions of the Z score of the outcome on the net present value of future sinking using perfect foresight expectations and a 6% discount rate. All regressions include borough by year and 1-km grid cell fixed effects. The hedonic index is the simple average of the Z-score of plot area, built area, parking spaces, elevators, bathrooms and bedrooms. Standard errors clustered at the census tract.

Figure A4: Residualized prices and sinking

(a) Residualized subsidence, 2010 (b) Residualized log appraisal values, 2010



Notes: Each point plots the residual from a regression of the outcome on property, borough by year, and vintage by year fixed effects. Residuals censored at the 1% and 99% percentiles and pseudo-log transformation applied to color scheme.

Table A1: Repeat sales sample compared to the full appraisal sample

	(1)	(2)	(3)	Mean Y
Single-family home	-0.166***	-0.151***	-0.067***	0.196
	(0.007)	(0.007)	(0.004)	
Apartment	0.171***	0.155***	0.070***	0.794
	(0.007)	(0.007)	(0.004)	
Empty lot	-0.001	-0.000	0.001	0.004
	(0.000)	(0.000)	(0.000)	
Communal property	-0.002***	-0.003***	-0.002***	0.004
	(0.001)	(0.001)	(0.000)	
Medium class	-0.005	-0.014	-0.015**	0.622
	(0.013)	(0.011)	(0.007)	
Plus class	-0.047***	-0.017***	-0.000	0.171
	(0.009)	(0.006)	(0.003)	
Luxury class	-0.020***	-0.013***	-0.002	0.035
	(0.002)	(0.002)	(0.001)	
Age	-5.518***	-5.196***	-2.191***	12.931
	(0.283)	(0.264)	(0.164)	
Plot size (sq. m)	122.636	150.028***	40.917	523.763
	(76.546)	(47.630)	(32.436)	
Construction size (sq. m)	-35.146***	-28.396***	-12.113***	97.059
	(1.344)	(1.353)	(0.645)	
N. floors	-0.237***	-0.199***	-0.205***	1.832
	(0.033)	(0.034)	(0.043)	
N. bedrooms	-0.241***	-0.212***	-0.096***	2.353
	(0.012)	(0.011)	(0.008)	
N. bathrooms	-0.309***	-0.219***	-0.077***	1.644
	(0.019)	(0.015)	(0.008)	
N. parking	-0.210***	-0.136***	-0.041***	1.129
	(0.021)	(0.018)	(0.010)	
Price (millions MXN)	-0.766***	-0.555***	-0.171***	2.224
	(0.049)	(0.044)	(0.015)	
Sinking to date (m)	-0.004	-0.038***	-0.014***	0.181
	(0.011)	(0.009)	(0.004)	
Rel. altitude (m)	0.193	0.130	-0.048	0.338
	(0.180)	(0.155)	(0.071)	
N. observations	322538	322538	322538	322538
Borough x Year FEs		X	X	
Census tract FEs			X	

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Each row is the coefficient of a separate regression where the outcome is listed in the row label and the explanatory variable is an indicator for being included in the repeat sales sample. All regressions include an intercept and cluster standard errors at the census tract level. The last column reports the overall mean of the outcome in each row.

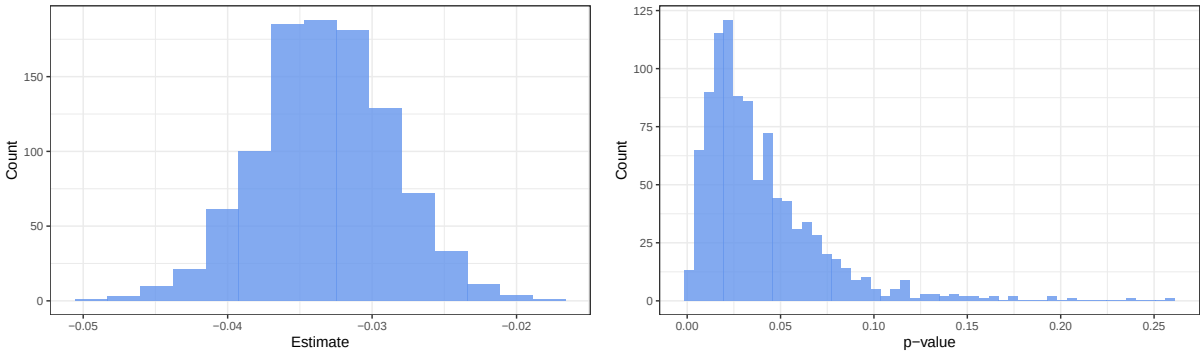
Table A2: Appraised values and transaction prices

	Price (millions MXN)		Absolute % deviation	
	(1)	(2)	(3)	(4)
Intercept		0.044***	0.038***	
		(0.007)	(0.005)	
Appraised value	0.967***	0.960***		
	(0.003)	(0.003)		
Sinking intensity (m)			0.001	0.000
			(0.003)	(0.003)
Observations	465,471	465,471	465,471	465,471
Mean Y	1.99	1.99	0.039	0.039
Adj. R2	0.497	0.497	0	0.014
Loan controls				X

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Columns (1) and (2) regress the transaction price (millions of pesos) on the appraised value in millions of pesos. The outcome in Columns (3) and (4) is the absolute value of the percent deviation in appraised values from transaction values, $|p_{transac} - p_{appraised}|/p_{transac}$. The explanatory variable in Columns (3) and (4) is average total sinking in the locality from 2007 to 2020; Column (4) includes controls for the income of the borrower, the interest rate assigned to the loan, bank-level fixed effects, and the size of the down payment. Standard errors in all specifications are clustered at the locality.

Figure A5: Monte Carlo estimates simulating measurement error in subsidence



(a) Distribution of Estimates of γ

(b) Distribution of p-values

Notes: Each plot shows the frequency of estimates of γ (Figure (a)) and its associated p-value (Figure (b)) given 1,000 draws of data in which each models total subsidence as $\tilde{S}_{jt} = S_{jt} + \sum_{i=0}^{Age_{jt}} e_{ij}$, where e_{ij} is drawn from a normal distribution with mean zero and standard deviation of 0.01 meters, simulating a 10 millimeter per year white noise measurement error associated with the measured sinking rate. Each point estimate is from a regression of log prices on $\tilde{S}_{jt}$, expected future sinking as measured with AR(1) expectations and a 6% discount rate, and property, vintage by year, and borough by year fixed effects. Estimation limited to properties with repeat sales. Standard errors clustered at the census tract.

Table A3: Effect of future sinking on rents

	AR(1)		Perfect foresight	
	3%	6%	3%	6%
Sinking to date (m)	-0.04630	-0.05000	-0.00625	-0.01237
	(0.18691)	(0.18240)	(0.17579)	(0.17181)
NPV future sinking	-0.00071	-0.00248	-0.00080	-0.00286
	(0.00265)	(0.00997)	(0.00107)	(0.00401)
log resident income	0.30590***	0.30589***	0.30553***	0.30550***
	(0.05282)	(0.05282)	(0.05267)	(0.05270)
N	671	671	671	671
Locality FEs	X	X	X	X
Year FEs	X	X	X	X
Age FEs	X	X	X	X
Type x Borough FEs	X	X	X	X
N. room x Borough FEs	X	X	X	X
N. bedroom x Borough FEs	X	X	X	X
Hedonic controls	X	X	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: The outcome in every column is the log rents. Sinking and expected future sinking are assigned using the confidential GPS location of the observation. Columns (1), (3) use a 3% discount rate to calculate the expected value of future sinking. Columne (2), (4) use a 6% discount rate to calculate the expected value of future sinking. Observations weighted by survey probability weights. All regressions include controls for log household income, an indicator for having a solid roof, an indicator for receiving water from the city every day, and fixed effects for locality, year, age, type (single family, apartment or other) by borough, number of rooms by borough, and number of bedrooms by borough. Standard errors clustered at the locality.

Table A4: Survey evidence on sorting and attention to sinking

	Surprised by severity (1)	Uses location to infer risk (2)	Any ex-ante factors used (3)
Expected 5-year sinking	0.169*	-0.102*	-0.031
	(0.093)	(0.058)	(0.063)
N	794	800	800
Mean Y (low sinking)	0.254	0.254	0.542
Mean X	0.284	0.284	0.284
Strata FEs	X	X	X
Respondent controls	X	X	X

* p <0.1, ** p <0.05, *** p <0.01

Notes: Outcomes in order of appearance are an indicator for reporting that sinking has been worse than expected (with omitted categories "better than expected" and "same as expected"), an indicator for using any information related to the location to infer future probability of having issues due to sinking, and an indicator for using any ex-ante factors (asking an expert, asking around, the news, the risk zone, examining construction quality, or neighborhood observation). "Expected 5-year sinking" is the average sinking rate in meters on the plot multiplied by five. All specifications include survey strata, respondent age, respondent gender, and respondent educational attainment fixed effects. All regressions weighted by sampling weights. Standard errors clustered at the census tract.

Table A5: Survey evidence for information frictions: Limiting to respondents who have moved in the last 10 years

	Surprised by severity (1)	Uses location to infer risk (2)	Any ex-ante factors used (3)
Expected 5-year sinking	-0.067	-0.297***	0.052
	(0.149)	(0.094)	(0.158)
N	156	158	158
Mean Y (low sinking)	0.283	0.156	0.528
Mean X	0.284	0.286	0.286
Strata FEs	X	X	X
Respondent controls	X	X	X

* p <0.1, ** p <0.05, *** p <0.01

Notes: Data limited to respondents who report moving to their current unit within the last 10 years. Outcomes in order of appearance are an indicator for reporting that sinking has been worse than expected (with omitted categories "better than expected" and "same as expected"), an indicator for using any information related to the location to infer future probability of having issues due to sinking, and an indicator for using any ex-ante factors (asking an expert, asking around, the news, the risk zone, examining construction quality, or neighborhood observation). "Expected 5-year sinking" is the average sinking rate times five. All specifications include survey strata, respondent age, respondent gender, and respondent educational attainment fixed effects. All regressions weighted by sampling weights. Standard errors clustered at the census tract.

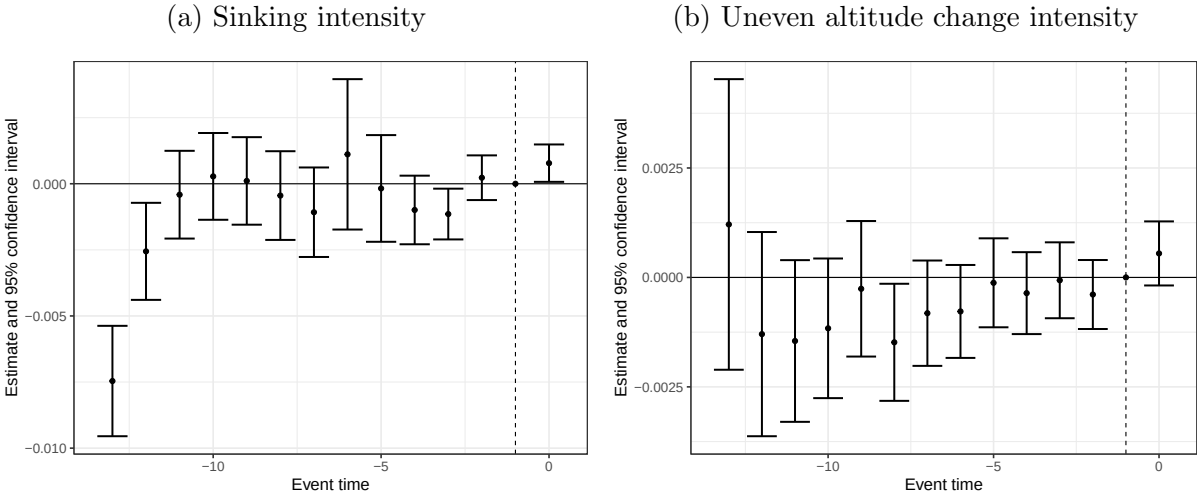
Table A6: Impact of sinking on the probability of new development in a pixel

	Pr. development on the Pixel		
	(1)	(2)	(3)
Pixel-level sinking rate (m/year)	0.00612***	0.00789***	0.00384***
	(0.00143)	(0.00148)	(0.00148)
Observations	1,491,345	1,491,345	1,491,345
Mean Y	0.005	0.005	0.005
Mean sinking rate (m/year)	0.053	0.053	0.053
Pixel FEs	X	X	X
Year FEs	X		
Zone x Year FEs		X	
Borough x year FEs			X

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The outcome in all regressions is an indicator for construction beginning on a new development in the pixel-year. All specification include 1-km grid cell fixed effects; (1) includes year fixed effects, (2) includes year by seismic zone fixed effects, and (3) includes borough by year fixed effects. Standard errors clustered at the census tract.

Figure A6: Pre-trends in pixel-level development



Notes: Depicts point estimates and 95% confidence intervals from a regression of an indicator for any development occurring on the pixel on an indicator for being a high-sinking pixel in 2020 (Panel (a)) or a highly-unevenly sinking pixel as measured by the cumulative relative altitude change (Panel (b)) in different periods. Periods denoted on the horizontal axis are the year relative to the first year that the pixel achieved a sinking intensity above the cut-off. Relative periods <10 are grouped into the -10 period. All regressions include pixel and year fixed effects. Point estimates calculated using Sun and Abraham (2021) staggered event study estimator. Standard errors clustered at the pixel.

Table A7: Robustness of cost of sinking estimates

	Main estimate: Conley SEs (1)	Additional controls (2)	Limiting to good appraisal areas (3)	Pre- 2017 earthquake (4)	Placebo: Neighbors' sinking (5)
Property-level sinking (m)	-0.04574**	-0.04671**	-0.08011**	-0.06448**	-0.04707**
	(0.01830)	(0.01952)	(0.03436)	(0.02522)	(0.01952)
NPV future sinking	0.00928	0.00146	-0.00346	-0.03815	0.00814
	(0.08998)	(0.09806)	(0.15052)	(0.12345)	(0.09679)
Sinking on neighboring properties (m)					0.00223
					(0.00638)
Observations	129,765	129,759	52,367	85,548	129,765
θ	-0.203	-0.031	0.043	0.592	-0.173
SE	(2.003)	(2.107)	(1.867)	(1.799)	(2.101)
ATE: plot sinking	-0.79%	-0.81%	-1.12%	-0.9%	-0.81%
Mean sinking (m)	0.173	0.173	0.14	0.14	0.173
Property FEs	X	X	X	X	X
Borough x Year FEs	X	X	X	X	X
Vintage x year FEs	X	X	X	X	X
N. bedrooms x time trends		X			
Type x time trends		X			
Binned surface area x time trends		X			

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: The outcome in all columns is the log appraised value. All estimates include property, vintage by year, and borough by year fixed effects. The perfect foresight measure of expected future sinking is used at 6% discounting. Estimation limited to properties with more than one appraisal. Column (1) presents estimates from our main specification, but with Conley standard errors with a 2-kilometer radius. Column (2) adds time trends by number of bedrooms, trends by type, and trends by quintile-binned area of construction. Column (3) limits to localities where the 75th percentile of differences between price and appraisal value is below median (0.08%). Column (4) limits to observations before September 2017. Column (5) includes as a regressor the average of cumulative sinking on neighboring pixels within a 100m radius of the appraisal from 2007 to the year of transaction. Standard errors clustered at the census tract in Columns (2)-(5).

Table A8: Impact of sinking intensity on firm outcomes

	Size		Sector					
	Employm. (1)	Sales (2)	Construc. (3)	Manuf. (4)	Wholesale (5)	Retail (6)	Services (7)	Public (8)
Sinking rate (m)	-1.472	-3.243	0.000	0.002	-0.001	-0.001	-0.001	0.000
	(1.532)	(2.830)	(0.000)	(0.002)	(0.001)	(0.004)	(0.003)	(0.002)
N	842,582	842,582	842,582	842,582	842,582	842,582	842,582	842,582
Mean Y	9.373	10.018	0.004	0.076	0.038	0.464	0.407	0.063
Census tract FEs	X	X	X	X	X	X	X	X
Borough x Year FEs	X	X	X	X	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes: Each observation is a firm-year. Outcomes in order of appearance are number of employees, sales (thousands MXM), and indicators for the firm being classified in the construction, manufacturing, wholesale, retail, services, or public sectors. All specifications include census tract, and borough by year fixed effects. Standard errors clustered at the census tract.

Table A9: Interest rates for mortgages are uncorrelated with tract-level sinking

	(1)	(2)
Intercept	10.066***	
	(0.038)	
Sinking intensity (m)	-0.039	-0.035
	(0.026)	(0.023)
Observations	465,471	465,471
Mean Y	10.043	10.043
Adj. R2	0	0.171
Loan controls		X

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Regression of the interest rate (in percentage points) charged on a mortgage on the total sinking in meters on the census tract from 2007 to 2020. Column (2) includes controls for the appraised value of the home, the borrower's income and the down payment as a share of the value, as well as bank and month fixed effects. Standard errors clustered at the census tract.

Table A10: Parameters needed for welfare analysis

	Value	**Interpretation**	**How estimated/calibrated**
γ	-0.046	Disutility from subsidence	Repeat transactions estimation
θ	0.5	Extent of pricing frictions	Value of future sinking in new home price
η	0.28	Supply elasticity	Redevelopment estimation
ρ	1/1.06	Discount factor	Chosen to reflect a 6% discount rate
μ	1.099	Mean of log cost shocks	Calibrated from distribution of structural residual in data
σ	0.379	Standard deviation of log cost shocks	Calibrated from distribution of structural residual in data
$\tilde{\xi}^S_{jt}$	(vector)	log cost shocks	Exactly matched to data using equilibrium conditions
e_{jt}	(vector)	Unobserved demand shifters	Exactly matched to data using equilibrium conditions
D	4.153	Demolition cost	Calibrated to match mean redevelopment probability of 0.45%
$\{\mathcal{S}_j\}$	(vector)	Distribution of sinking shocks	AR(1)
$\overline{L}$	6,227,549	Population	Total number of households in Mexico City plus Estado de México in 2010 census
$\{H_{j0}\}$	(vector)	Housing density	Calibrated from 2010 census

Table A11: Pigouvian tax on groundwater pumping and water consumption, rates

	Very low income	Low income	Middle income	Unsubsidized
Share of HHs	0.243	0.42	0.113	0.225
Avg. consumption (m^3)	23.35	29.427	27.294	35.098
Avg. marginal rate	\$0.303	\$0.574	\$1.167	\$1.691
Average bi-monthly bill	\$4.175	\$13.154	\$22.106	\$59.34
Tax as % of marginal rate	968.74%	510.93%	251.38%	173.48%
Tax as % of bill	984.46%	393.79%	217.33%	104.11%

Notes: Rates from the 2024 schedule. Average consumption estimated from city block averages reported in 2019. All monetary values converted to 2018 USD using a 19 MXN/USD exchange rate, and deflating 2024 prices to 2018. "Marginal rates" refer to the marginal price of water at average city block consumption per household.

A1 Data Appendix

A1.1 Subsidence data

Subsidence calculated from Sentinel-1 measures (2014-2020) at the pixel level (90 by 90 meter resolution) using the C-band was directly provided by Chaussard et al. (2021). I create an annual pixel-level measure of sinking by mimicking their methodology for creating the average pixel-level analysis reported in their paper by estimating a linear regression of cumulative vertical deformation on time for each pixel-year, the slope of which is an estimate of the sinking in that period. This method is preferred to directly integrating sinking at each time step, which is sensitive to estimates contaminated by error or low coherence (Chaussard et al., 2021). In our setting the linear projection helps remove measurement error that may be related to construction itself, as a change in building height, if detected at all by the satellite, would be smoothed over by the other measurements.

We create annual, pixel-level measures of subsidence for 2007-2011 using ALOS-1 ascending track L-band SAR acquisitions for the Mexico Valley. We process the raw Level-1 data using the ISCE full stack processing framework (Interferometric synthetic aperture radar Scientific Computing Environment) and use MintPy algorithms to correct for tropospheric distortions and unwrapping errors and produce the final corrected time series values (Yunjun et al., 2019) and GACOS to do tropospheric corrections (Yu et al., 2018b,a, 2017). Given the high signal-to-noise ratio in sinking in central Mexico, I do not filter on coherence to maximize the coverage of the data.

Using the same process, we create annual, pixel-level measures of subsidence for 2021-2024 using Sentinel-1 ascending track C-band SAR acquisitions for the Mexico Valley. We process the raw Level-1 data using the ISCE full stack processing framework (Interferometric synthetic aperture radar Scientific Computing Environment) and use MintPy algorithms to correct for tropospheric distortions (using GACOS) and unwrapping errors. I do not filter on coherence. Both this series and the ALOS-1 series are converted from line-of-sight (LOS) measures to vertical displacement assuming no horizontal movement, so that $VERT = LOS/cos(incidenceangle)$, in keeping with (Chaussard et al., 2021).

In order to create a harmonized panel of pixels, I create a regular 100m$\times$100m pixel overlay that spans the two spatial extents of the ALOS and Sentinel data and take an intersection-weighted average of the original pixel subsidence rates. From this grid I interpolate linearly from 2011-2013.

A1.1.1 Measurement error

In the direct measures of subsidence produced by SAR interferometry, measurement error arises from four main sources: (i) errors in the characterization of the exact orbit location, which is required to properly compare multiple passes; (ii) errors arising from troposhperic or ionospheric conditions that affect measurement, in particular water vapor; (iii) unwrapping errors, which occur when "unwrapping" the phase length which is modular 2π, and (iv) temporal decorrelation, where measurements jump in the 2π range due to irregular backscatter. We discuss the role for these in our measurements and the magnitude of errors from these sources.

Orbital location errors Errors in characterizing the exact location of the satellite in space at each pass can result in long wavelength phase ramps which essentially represent a level shift in subsidence that is common to every pixel. These are relatively straightforward to identify in interferograms, and the MintPy algorithm we use to generate the time series implements a linear "de-ramping" of the time series. That said, before de-ramping, the accuracy of orbit information depends on the satellite. The Sentinel-1 orbit information, used in our analysis from 2014-2024, is extremely accurate, and Sentinel's orbit products have an accuracy typically below 5cm in 3D in terms of the exact location of the satellite (Peter et al., 2017). This results in negligible orbit errors in the resulting interferometry process. As an older satellite, the ALOS-1 satellite achieves an accuracty of around 40 centimeters in 3D in terms of the satellite's location in space, so de-ramping is more important for this satellite product.

Tropospheric and ionospheric errors Changes in tropospheric and ionospheric conditions can distort interferograms, which manifests as a phase ramp similar to orbital location errors. We correct for differences in the tropophere by using GACOS hourly temperature, pressure, and water vapor measurements to adjust measurements using MintPy. The magnitude of ionospheric distortions is inversely proportional to the band frequency; in the case of the C-band, which is relatively high frequency, these errors are minimal, on the order of <5 millimeters. For the lower frequency L-band, the ionosphere plays a mroe important role, but its importance is highly dependent on context; ionospheric errors are at their worst during periods of high solar activity, during the day at equitorial and polar locations. The ALOS-1 satellite has a sun-synchronous orbit, so it's ascending orbit passed over Mexico City around 10:30PM local time. The nighttime pass together with the fact that Mexico City is near mid-latitude ranges at 19° latitude means that ionospheric errors are somewhat mitigated, and our de-ramping also decreases ionospheric delay (Chaussard et al., 2021).

Unwrapping errors Unwrapping errors result from too large of a phase difference between passes, and rather than affecting measurements directly it affects the ability to produce a measurement at all in some places. Unwrapping errors can arise from several sources; those most relevant to our setting include (i) too fast of ground deformation; and (ii) steep slopes that mean that measurement is sensitive to the exact line-of-sight of the satellite. Our area of interest does not have particularly steep slopes, as urbanized Mexico City is located within a valley. How fast is "too-fast" of deformation depends on the band, as the phase is wrapped around half the radar's wavelength. For the Sentinel-1 C-band, this means that if displacement exceeds 2.8 centimeters between passes, which occur every 6 days, then displacement would not be able to be estimated. Thus, subsidence would need to exceed a rate of 1.7 meters per year to be undetectable, a rate observed for dramatic events such sudden, discrete events such as volcanic eruptions or ice sheet collapses, but not observed in our sample. The L-band measures we use from the ALOS-1 satellite are able to detect up to 11.8 centimeters between passes, and our average pass frequency is 90 days, so sinking would have to exceed 47.9 centimeters per year to be undetectable, which is almost double the maximum rate observed in our data.

Temporal decorrelation The primary source of temporal decorrelation is changes in the ground surface that cause measurements to behave differently between passes, for example changes in vegetative density or snow cover. These sources of error are minimized in

urban settings, where concrete provides a time-consistent radar signature. This is diagnosable using the average coherence in the signal between satellite passes, which when limiting to Mexico City in our data is over 0.9 on a scale from 0 to 1 almost everywhere.

Summing up By focusing on an urban setting and using state-of-the-art tools for conducting interferometry with tropospheric corrections and linear de-ramping, we minimize the role for measurement error in our analysis. Other studies in urban areas use GPS stations that provide ground measures at discrete points, and find measurement errors on the order of less than millimeters per year (see: 1.5mm/year in Beijing (Liu et al., 2020); 9mm/year in the Antelope Valley, CA (Bawden et al., 2023)). Assuming we achieve similar accuracy to the Beijing study (1.5mm), as this more closely mirrors our urban setting, our measurement error would be an order of magnitude of around 2.1% of the average sinking rate of 7 centimeters/year.

A1.1.2 Does development cause subsidence?

While historic buildings in Mexico City such as the Palace of Fine Arts have been known to cause sinking because of their own weight, modern construction code requirements seek to prevent this through soil studies, including underground levels that offset the weight of the building, and using pylons that "anchor" the building into the ground instead of sitting on top of the soil.

We test whether development causes subsidence by estimating the following event study specification:

$$s_{pt} = \alpha_p + \alpha_{g(p)t} + \sum_{\tau \neq -1} \beta^{\tau} 1(\text{New development}_i) \times 1(t_p^* - t = \tau) + e_{pt} \tag{1}$$

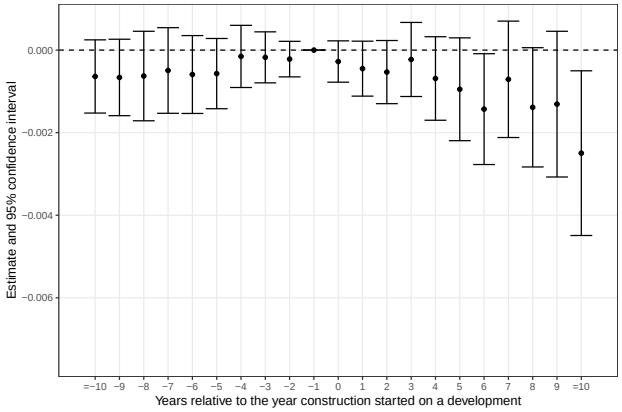
where s_{pt} is the sinking rate on pixel p in period t and $1(\text{New development}_i)$ is an indicator for that pixel ever having a development in our sample period. t_p^* is the year that construction started on a development in that pixel. We also control for pixel fixed effects α_p and borough by year fixed effects $\alpha_{g(p)t}$. To avoid confounding the effect of multiple developments, we limit estimation to the sample of pixels that had either one or no developments over the period. Standard errors are clustered at the census tract.

Figure AA1 reports $\widehat{\beta^{\tau}}$. There is no systematic trend in sinking before or after development, and taking point estimates at face value would imply that a new development at most causes an average fall of 1 milimeter in the sinking rate, representing approximately a 4% slowing in sinking. This is generally consistent with studies from Mexico City and elsewhere that find that the role of human development in causing land subsidence is minor (Chaussard et al., 2021; Parsons et al., 2023). The fact that the coefficients are small and largely insignificant also assuages concerns that construction impedes measurement of subsidence by masking as positive elevation gain.

A1.2 Survey design and implementation

The survey was designed to be representative of Mexico City housing units, with stratification on two broad zones of high (7 boroughs) and low (9 boroughs) sinking from 2007-2020 defined

Figure AA1: Impact of development on future sinking rates



Notes: Plots $\widehat{\beta}^{\tau}$ and 95% confidence interval from estimating Equation (1). Periods ≤ -10 are grouped into a single category as are periods $\geq +10$. Standard errors clustered at the census tract.

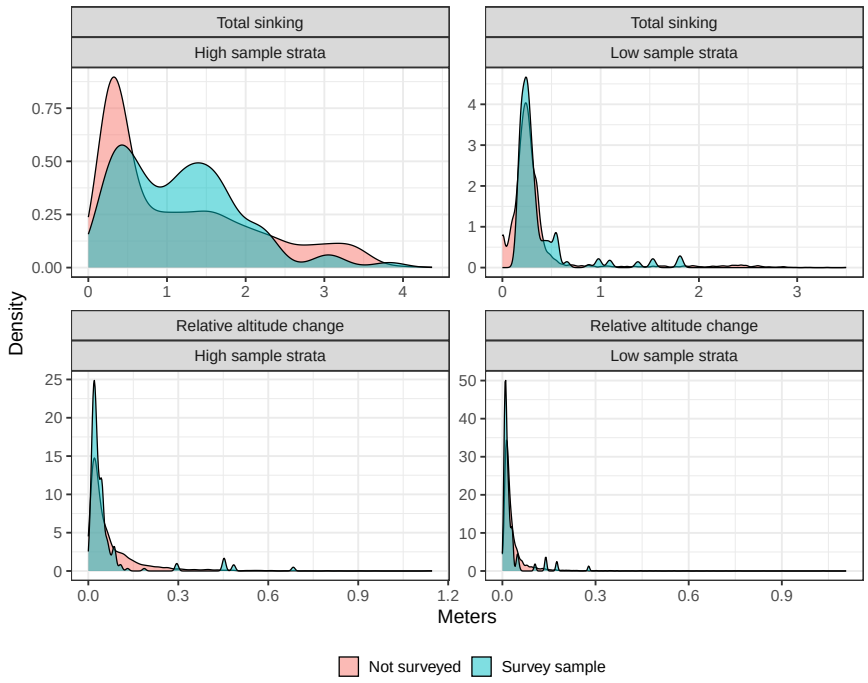
at the borough level. Within strata, census tracts were sampled with probabilities drawn from the number of housing units in that tract in the 2020 census; 80 tracts were randomly sampled, and 10 households were surveyed per tract. IRB exemption was obtained through the UC Berkeley Office for the Protection of Human Subjects.

The surveyed homes provide good representation of sinking across the city; Figure AA2 reports the distribution of average total sinking and total relative altitude loss from 2008-2020 for surveyed and non-surveyed tracts. The distributions are very similar, with some more mass in the surveyed tracts at mid-levels of sinking coming from the distribution of housing units over space within high-risk boroughs.

Table AA1 reports comparisons of our resident and unit outcomes with the 2022 Income and Expenditure Survey (ENIGH). Despite the three-year difference and different sampling strategy (housing units versus residents), outcomes are broadly comparable though statistically differentiable between our survey and the ENIGH. Given the sample restriction on homeowners of any tenure and renters with at least 5 years in the unit, our sample is 73% homeowners (compare to 60% in the ENIGH). Looking within homeowners in Table AA1, we see that survey respondents are more likely to inhabit single-family homes and live in self-built homes as compared to commercially built homes. Respondents are also significantly older on average. These differences likely highlight selection into being contactable by the survey firm. Reassuringly, our respondents are not meaningfully more educated on average; a 0.58 difference in educational attainment represents an approximate difference of 0.87 years of schooling.[1]

[1]Schooling is coded as a numeric factor representing categories such as "Middle school incomplete", "Middle school complete", "High school incomplete", etc. Given that the modal respondent completed at least middle school and that middle and high school in Mexico each last 3 years, a each unit increase in the education variable represents approximately 1.5 years of schooling.

Figure AA2: Distribution of subsidence in surveyed and non-surveyed census tracts



Notes: Each plot is the distribution of average total sinking from 2008-2020 at the census tract level and the total relative altitude change from 2008-2020 on surveyed and non-surveyed tracts, reported separately for the high-sinking strata (N=500) and the low-sinking strata (N=300).

Table AA1: Balance of survey outcomes against the 2022 ENIGH

		Full sample		Owners only			
		ENIGH	Our survey	ENIGH	Our survey	Diff, full sample	Diff, owners sample
Unit characteristics							
Single-family home	Mean	0.66	0.78	0.70	0.79	0.12***	0.09***
	SD	(0.437)	(0.412)	(0.409)	(0.401)	(0.019)	(0.022)
	N	2541	800	1595	581	3341	2176
Apartment	Mean	0.31	0.22	0.27	0.21	-0.09***	-0.06***
	SD	(0.422)	(0.412)	(0.392)	(0.401)	(0.019)	(0.022)
	N	2541	800	1595	581	3341	2176
Age of home	Mean	26.82	38.82	30.39	42.28	4.5***	8.27***
	SD	(20.285)	(22.713)	(19.471)	(22.387)	(0.96)	(1.106)
	N	2092	800	1471	581	2892	2052
Self-built	Mean	0.28	0.63	0.47	0.67	0.16***	0.2***
	SD	(0.49)	(0.483)	(0.49)	(0.47)	(0.023)	(0.025)
	N	1597	800	1597	581	2397	2178
Commercially built	Mean	0.26	0.22	0.44	0.23	-0.22***	-0.2***
	SD	(0.465)	(0.409)	(0.465)	(0.42)	(0.021)	(0.023)
	N	1597	800	1597	581	2397	2178
Respondent characteristics							
Female	Mean	0.54	0.57	0.54	0.56	0.04***	0.02***
	SD	(0.499)	(0.496)	(0.499)	(0.498)	(0.019)	(0.022)
	N	8421	800	5449	581	9221	6030
Age (respondent)	Mean	38.45	52.94	41.63	53.73	14.55***	12.11***
	SD	(21.547)	(16.762)	(22.135)	(17.086)	(0.671)	(0.818)
	N	8421	800	5449	581	9221	6030
Education	Mean	4.23	6.12	4.44	6.24	0.71***	0.58***
	SD	(3.541)	(3.081)	(3.563)	(3.118)	(0.123)	(0.148)
	N	6505	800	4221	581	7305	4802

Notes: Means and standard deviations are weighed by sample weights to the population. Differences are conditional on strata fixed effects and use heteroskedasticity-robust standard errors. Stars represent p-values less than 0.10 ($*$), 0.05 ($**$), and less than 0.01 ($***$).

A1.3 Appraisal data

A1.3.1 Appraised useful life remaining

We investigate whether appraised values for the years of useful life remaining follow systematic rules that would preclude us from exploiting true variation in the home's physical state. If appraisers use fixed rules based on some property characteristic or age to assess the useful life left, then regressions that include property and year of appraisal by year of last appraisal fixed effects would control for all possible variation in this variable.

To investigate this, we define the implied duration of the property's useful lifespan by adding the age at time of appraisal to the appraised years of useful life remaining. Then, we investigate the extent to which the modal value of these implied useful lifespans by the property's borough, type (basic, medium, luxury amenities), vintage, and year of appraisal explains the appraised useful lifespan. We find that there are systematic patterns in the modal appraised useful life, with different boroughs and vintages receiving common appraised useful lifespans of 65, 70, 75, or 80 years. However, there are also substantial deviations from these rules of thumb.

We compute the cell-specific modal useful lifespan for cells defined by the property's location, type, vintage, and year of appraisal, and plot the deviation of the implied useful lifespan from this modal value in Figure AA3. While differences are centered around zero, there are appraised useful lifespans that differ substantially from the rule of thumb value. If we define the rule of thumb excluding vintage, we find that newer homes appear to be built with longer useful lives, a finding consistent with technological improvements over time that increase the durability of homes.

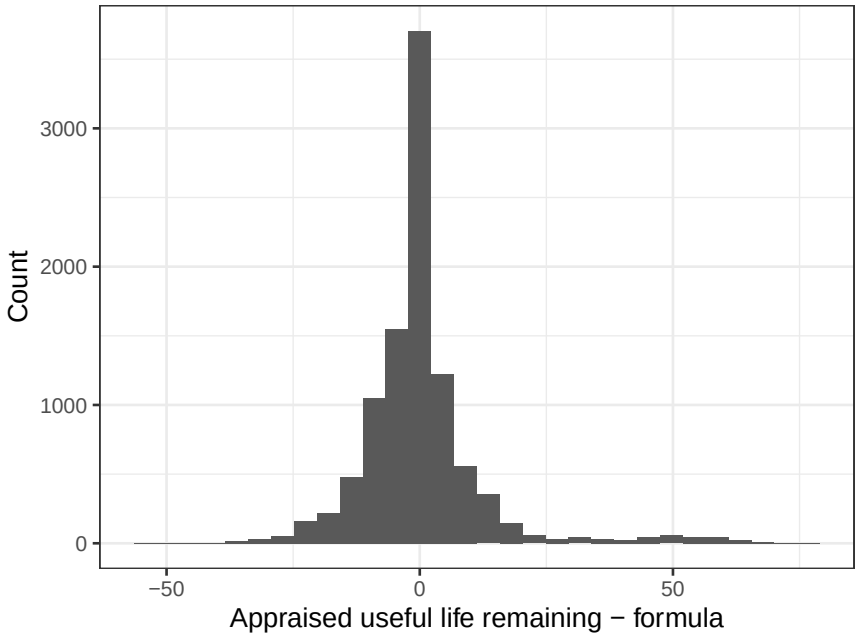
A1.4 Housing developer data

We aggregate the quarterly development panel into a dataset of developments by calculating averages of characteristics of the modal unit and its asking price that are weighted by the number of units reported sold in that period. To create a dataset of new units developed, we expand the data by the number of units sold, assigning to each the modal price and characteristics reported in that quarter.

A1.4.1 Coverage of the new housing market

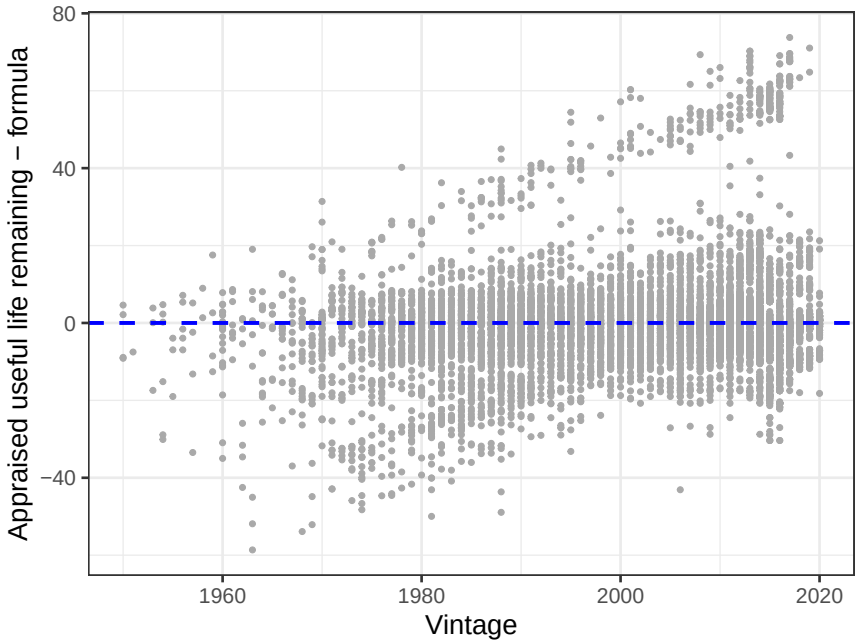
Across Mexico, the Central Bank reports that the DIME should cover about 60% of new homes based on the fact that developers account for 60% of new unit construction. We compare the units reported built in the DIME to the number of appraisals per year to get a better idea of the DIME coverage in Figure AA5. Units reported sold by the DIME surpass even the total number of appraisals in some years, and is consistently around double the number of appraisals for new builds (blue line). This is likely because the appraisal data is missing transactions that did not involve a mortgage, a significant share (around 70%). While conclusions about coverage should be tempered by this fact, we see this as suggestive that the DIME covers a significant share of new development in Mexico City.

Figure AA3: Deviations of appraised useful lifespan from rule of thumb values



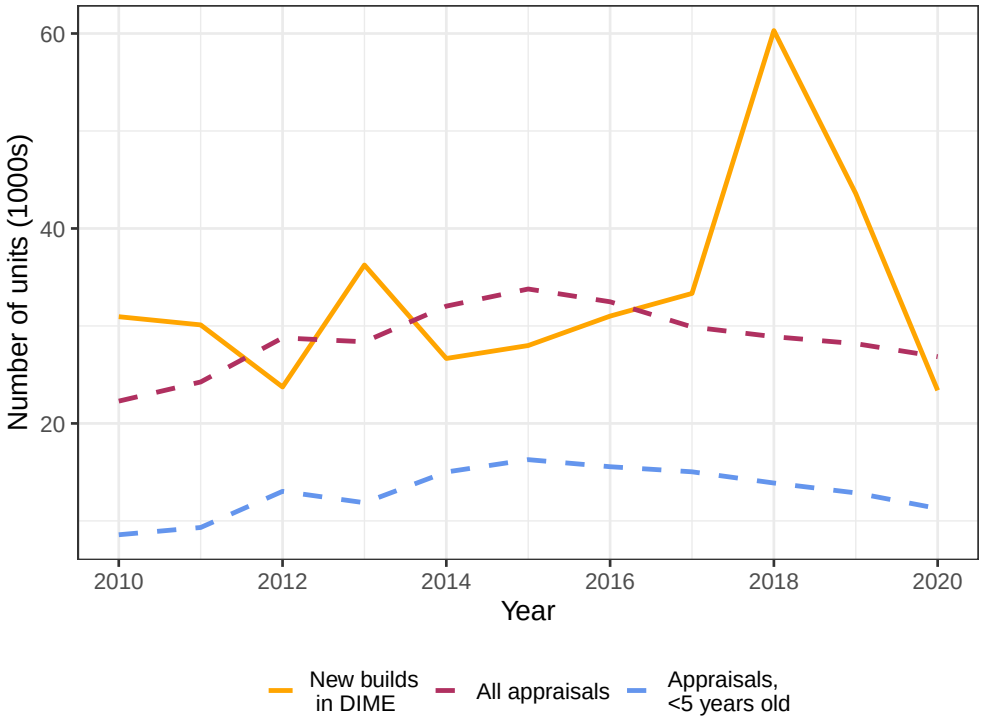
Notes: Histogram of the appraised useful lifespan minus the implied rule of thumb using a 10,000-observation random sample from the data. The rule of thumb is defined as the modal appraised useful lifespan for properties that are located in the same borough, built in the same year, appraised in the same year, and are of the same type.

Figure AA4: Deviations of appraised useful lifespan from rule of thumb values



Notes: The rule of thumb is defined as the modal appraised useful lifespan for properties that are located in the same borough, appraised in the same year, and are of the same type, but does not control for vintage.

Figure AA5: Total number of newly developed units and appraisals



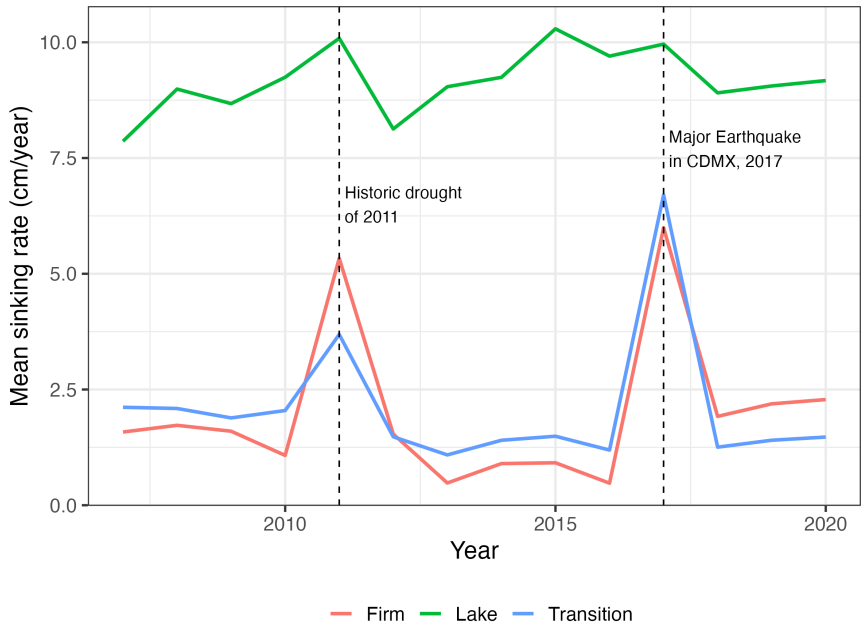
Plots the total number of units per year. The solid orange line plots the total number of units reported sold by the DIME. The dashed lines represent the total number of appraisals (maroon) and the total number of appraisals that are reported less than 5 years since construction (blue).

A2 Predictability of subsidence over time and space

As seen in Figure 2, subsidence intensity is very serially correlated over space, with the fastest rates of sinking concentrated over the historic lake and the surrounding "transition" zone defined in the construction code. Subsidence is also highly serially correlated; at the pixel level, the lagged subsidence rate has a correlation of 0.85 with the contemporaneous subsidence rate.

However, how sinking has evolved over time varies substantially in Mexico City. First, how sinking rates respond to shocks that affect groundwater levels or geophysical stability depends on zone. Figure AA6 plots subsidence rates over time by broad categories of the geophysical environment defined in the construction code. While subsidence rates are consistently high in the lake zone, the more resistant Firm and Transition zones are more responsive to shocks; each experiences a spike in subsidence rates during the drought of 2011 and the large earthquake of 2017. In particular after the earthquake, rates in the Firm zone experienced a level shift up in the sinking rate.

Figure AA6: Subsidence rates over time



Notes: Lines represent simple averages of subsidence rates by year and seismic zone as defined by the construction code.

A3 Other explanations for low capitalization

The literature on mispricing of other environmental hazards has found important roles for both behavioral factors such as myopic behavior (Gallagher, 2014), information frictions (Bakkensen and Barrage, 2022; Hino and Burke, 2021), and inattentiveness absent large realizations (Bakkensen et al., 2019; Ortega and Taşpınar, 2018; Petkov and Ortega, 2025)–

factors that may be even more important in emerging economy housing markets where disclosure and other mandatory quality reporting are not present. Policy can also play a role, and subsidized insurance (Bézy, 2025; Wagner, 2022) or bailout programs may act to effectively depress expectations about future costs, though we note the absence of such policies in this context.

One explanation is that homebuyers have very high discount rates or are very risk-loving when it comes to subsidence. While the levels of discounting and risk aversion measured by our survey are unlikely to be directly comparable to how households might discount the stream of housing services,[2] we can use these measures to test whether households that more heavily discount the future tend to sort into sinking areas. Table AA2 presents results; neither discount rates nor risk aversion are correlated with sinking intensity or realized sinking, suggesting these types of preferences do not drive results.

Table AA2: Subsidence and sorting on risk, discounting preferences

	Discounting		Risk aversion	
	(1)	(2)	(3)	(4)
Expected 5-year sinking	-0.184		0.264	
	(0.300)		(0.170)	
Sinking in last 5 years (m)		-0.084		0.184
		(0.318)		(0.159)
N	396	396	362	362
Mean Y (low sinking)	0.671	0.671	0.624	0.624
Mean X	0.271	0.294	0.323	0.353
Strata FEs	X	X	X	X
Respondent controls	X	X	X	X

* p <0.1, ** p <0.05, *** p <0.01

Notes: Outcomes in order of appearance are the implied discount rate and risk aversion coefficient measured by Falk instruments (higher discount rates indicate higher impatience). Respondents had a random 50% change on being asked the discounting or risk aversion questions, with 400 being surveyed on each each. "Expected 5-year sinking" is the average sinking rate on the property over the full panel multiplied by five. "Sinking in last 5 years (m)" is the total subsidence on the property over the last 5 years. All specifications include survey strata, borough, respondent age, respondent gender, and respondent educational attainment fixed effects. All regressions weighted by sampling weights. Standard errors clustered at the census tract.

[2]Following Falk et al. (2016), we measure discount rates by asking households a series of questions that consider delaying $29USD (500 MXN) over 12 months. Risk aversion is measured by asking households to choose between a 50-50 lottery to win $88 USD ($1,500 MXN) and a series of risk-free prizes. Also note that discount rates for housing should be tied to the discounting of future values to other financial products, which further minimizes the theoretical role for subjective discounting.

A4 Theory Appendix

A4.1 Linearization of expectation terms

First, consider a first-order approximation of $g(S_{j\tau}^{t+1}) = \exp(\gamma S_{j\tau}^{t+1})$ around $S_{j\tau}^{t+1} = \mathbb{E}(S_{j\tau}^{t+1})$:

$$\begin{aligned} g(S_{j\tau}^{t+1}) &= \exp(\gamma \mathbb{E}(S_{j\tau}^{t+1})) + \gamma \exp(\gamma \mathbb{E}(S_{j\tau}^{t+1})) \times (S_{j\tau}^{t+1} - \mathbb{E}(S_{j\tau}^{t+1})) + R(S_{j\tau}^{t+1}) \\ &\Rightarrow \mathbb{E}(g(S_{j\tau}^{t+1})) = \exp(\gamma \mathbb{E}(S_{j\tau}^{t+1})) + \mathbb{E}(R(S_{j\tau}^{t+1})) \end{aligned}$$

The remainder term $R(S_{j\tau}^{t+1})$ is a function of the sum of higher-order moments, each of size:

$$\frac{\gamma^n}{n!}(S_{j\tau}^{t+1} - \mathbb{E}_t(S_{j\tau}^{t+1}))^n$$

These terms will be small if the moments of the distribution of sinking grow at a slower rate than $\frac{\gamma^n}{n!}$. As an example, we can examine the second-order approximation, which include the variance:

$$\begin{aligned} g(S_{j\tau}^{t+1}) \approx & \exp(\gamma \mathbb{E}(S_{j\tau}^{t+1})) + \gamma \exp(\gamma \mathbb{E}(S_{j\tau}^{t+1}))(S_{j\tau}^{t+1} - \mathbb{E}(S_{j\tau}^{t+1})) \\ & + \frac{\gamma^2}{2} \exp(\gamma \mathbb{E}(S_{j\tau}^{t+1}))(S_{j\tau}^{t+1} - \mathbb{E}(S_{j\tau}^{t+1}))^2 \\ & \Rightarrow \mathbb{E}(g(S_{j\tau}^{t+1})) \approx \exp(\gamma \mathbb{E}(S_{j\tau}^{t+1})) + \frac{\gamma^2}{2} \exp(\gamma \mathbb{E}_t(S_{j\tau}^{t+1})) Var(S_{j\tau}^{t+1}) \end{aligned}$$

This is small in the time series of plot-level sinking.

Our second step, after implementing the above approximation, us to linearize the log term around these expectations. Consider a first-order Taylor series approximation of:

$$\begin{aligned} & f(\mathbb{E}_t(\{\tilde{\lambda}_{ju}\}), \mathbb{E}_t(\{e_{ju}\}), \mathbb{E}_t(\{s_{ju}\}_u)) = \\ & \log\left(\exp(-\tilde{\lambda}_{jt} + e_{jt}) + \sum_{\tau=t+1}^{\infty} \rho^{\tau-t} \exp(\gamma \mathbb{E}_t\{S_{j\tau}^{t+1}\} - \mathbb{E}_t(\tilde{\lambda}_{j\tau}) + \mathbb{E}_t(e_{j\tau}))\right) \\ & = \log\left(\exp(-\tilde{\lambda}_{jt} + e_{jt}) + \sum_{\tau=t+1}^{\infty} \rho^{\tau-t} \exp(-\mathbb{E}_t(\tilde{\lambda}_{j\tau}) + \mathbb{E}_t(e_{j\tau}) + \sum_{u=t+1}^{\tau} \gamma \mathbb{E}_t\{s_{ju}\})\right) \end{aligned}$$

around $\tilde{\mathbb{E}}_t(s_{ju}) = 0$, $\mathbb{E}_t(e_{ju}) = e_{jt}$, and $\mathbb{E}_t(\tilde{\lambda}_{ju}) = \lambda_{jt}$ $\forall u > t$. Note this is analogous to assuming that these variables follow a random walk. In the case of e_{jt} this reduces to assuming a distribution for these shocks. In the case of the $\tilde{\lambda}_{jt}$ this approximation comes at a cost in terms of model consistency, as these relative choice probabilities are equilibrium objects that have a distribution that is not likely to be a random walk.[3] However, this approximation reflects the fact that $\tilde{\lambda}_{j\tau}$ is fixed until re-development; because re-development is rare, with a per-period probability of 2.7%, $\mathbb{E}_t(\tilde{\lambda}_{j\tau}) = Pr(d_{jt} = 0)\lambda_{jt} +$

[3]This distribution is difficult to characterize for a given j, as it is a convolution of the distribution of cost shocks and a non-linear function of the random variables in demand that affect prices.

$Pr(d_{jt}=1)\mathbb{E}_t(\tilde{\lambda}_{j\tau}|d_{jt}=1) \approx \lambda_{jt}$.

$$\begin{aligned} f(\{\mathbb{E}_t(\tilde{\lambda}_{ju})\},\{\mathbb{E}_t(e_{ju})\},\{\mathbb{E}_t(s_{ju})\}) \approx & f(\tilde{\lambda}_{jt}, e_{jt}, \vec{0}) \\ & + \sum_{u=t+1}^{\infty} \frac{\partial f}{\partial \tilde{\lambda}_{ju}}\Big|_{\tilde{\lambda}_{jt},e_{jt},\vec{0}} \times (\mathbb{E}_t(\tilde{\lambda}_{ju}) - \lambda_{jt}) \\ & + \sum_{u=t+1}^{\infty} \frac{\partial f}{\partial e_{ju}}\Big|_{\tilde{\lambda}_{jt},e_{jt},\vec{0}} \times (\mathbb{E}_t(e_{ju}) - e_{jt}) \\ & + \sum_{u=t+1}^{\infty} \frac{\partial f}{\partial \mathbb{E}_t(s_{ju})}\Big|_{\tilde{\lambda}_{jt},e_{jt},\vec{0}} \times (\mathbb{E}_t(s_{ju}) - 0) \end{aligned}$$

Note that:

$$\begin{aligned} f(\tilde{\lambda}_{jt}, e_{jt}, \vec{0}) &= \log\left(\frac{\exp(e_{jt}-\tilde{\lambda}_{jt})}{1-\rho}\right) \\ \frac{\partial f}{\partial \tilde{\lambda}_{ju}}\Big|_{\tilde{\lambda}_{jt},e_{jt},\vec{0}} &= -\left(\frac{1-\rho}{\exp(e_{jt}-\tilde{\lambda}_{jt})}\right)\rho^{u-t}\exp(e_{jt}-\tilde{\lambda}_{jt}) = -(1-\rho)\rho^{u-t} \\ \frac{\partial f}{\partial e_{ju}}\Big|_{\tilde{\lambda}_{jt},e_{jt},\vec{0}} &= \left(\frac{1-\rho}{\exp(e_{jt}-\tilde{\lambda}_{jt})}\right)\exp(e_{jt}-\tilde{\lambda}_{jt})\rho^{u-t} = (1-\rho)\rho^{u-t} \\ \frac{\partial f}{\partial \tilde{\mathbb{E}}_t(s_{ju})}\Big|_{\tilde{\lambda}_{jt},e_{jt},\vec{0}} &= \left(\frac{1-\rho}{\exp(e_{jt}-\tilde{\lambda}_{jt})}\right) \times \exp(e_{jt}-\tilde{\lambda}_{jt})\frac{\gamma\rho^{u-t}}{1-\rho} = \gamma\rho^{u-t} \end{aligned}$$

The last line follows from the fact that a given $\mathbb{E}_t(s_{ju})$ term appears in every (discounted) term starting in period u to infinity.

Thus:

$$\begin{aligned} f(\{\mathbb{E}_t(\tilde{\lambda}_{ju})\},\{\mathbb{E}_t(e_{ju})\},\{\tilde{\mathbb{E}}_t(s_{ju})\}) \approx & \; e_{jt} - \tilde{\lambda}_{jt} - \log(1-\rho) \\ & + (1-\rho)\sum_{\tau=t+1}^{\infty}\rho^{\tau-t}\left(\mathbb{E}_t(e_{j\tau}) - e_{jt} - (\mathbb{E}_t(\tilde{\lambda}_{j\tau}) - \tilde{\lambda}_{jt})\right) \\ & + \gamma\sum_{\tau=t+1}^{\infty}\rho^{\tau-t}\tilde{\mathbb{E}}_t(s_{j\tau}) \end{aligned}$$

A4.2 Calculating expectations of future sinking

We create two measures of expectations of future sinking. The first assumes that sinking rates follow an AR(1) process at the plot level, such that:

$$s_{jt+1} = \mu_j + \phi s_{jt} + e_{jt+1} \tag{2}$$

where $\mu_j = (1-\phi)\overline{s}_j$ We estimate the parameters of this function, namely, $\{\mu_j\}_j$ and ϕ, and

define the expected value of the sinking rate in each period as:

$$\mathbb{E}_t(s_{jt+\tau}) = \mu_j(1-\phi^{\tau-t}) + \phi^{\tau-t}s_{jt} \tag{3}$$

with the expected value of total sinking in τ being the sum of the expectations of sinking rates from $t+1$ to τ. Solving for the net present value of these terms from $t+1$ to infinity gives:

$$\sum_{\tau=t+1}^{\infty} \rho^{\tau-t}\mathbb{E}_t(s_{j\tau}) = \frac{\rho}{1-\rho}\mu_j + \frac{\rho\phi}{1-\rho\phi}(s_{jt}-\mu_j) \tag{4}$$

For expectations that follow perfect foresight, we use the realized sinking on the plot from $t+1$ through 2024, and use a plot-specific estimate of the rate from 2025-on for sinking after that date. The expectation term under perfect foresight is:

$$\sum_{\tau=t+1}^{\infty} \rho^{\tau-t}\mathbb{E}_t(s_{j\tau}) = \sum_{\tau=t+1}^{2024} \rho^{\tau-t}s_{j\tau} + \frac{\rho^{2025-t}}{1-\rho}\widehat{s}_j \tag{5}$$

A4.3 Welfare calculations

A4.3.1 Renters

Per-period surplus per renter in the housing unit market is:

$$CS_j^i = R_j(0)\log\left(\sum_k \exp\{-\log(R_k(0)) + Z_k\delta + e_k\}\right) + C$$

In this expression, the marginal utility of money (rent, evaluated in the initial period) is $\frac{\partial V_j}{\partial R_j(0)} = -1/R_j(0)$. We measure the welfare loss to consumers Δ^D as the equivalence variation required to compensate home buyers for the impacts of subsidence relative to the benchmark. Because of the size of the system, it is too computationally intensive to calculate an equivalent variation for each consumer, so we calculate the average equivalent variation as:

Δ_t^D such that:

$$\frac{1}{J}\sum_j CS_{jt}^i = \frac{1}{J}\sum_j R_j(0)\log\left(\sum_j \exp\{-\log(R_j(0)+\Delta_t^D) + Z_j\delta + e_{jt}\}\right) \tag{6}$$

While a building is not redeveloped, rents must adjust so that equilibrium demand on the plot is equal to the (fixed) supply of housing on the plot, so we can express mean utility in each period as a function of utility in the period when the building is new ($S_{jt}=0$) with rental price $R_j^*(0)$.

A4.3.2 Property managers

Expected manager surplus is their profits under rational expectations:

$$\Pi_t = \sum_j H_{jt} \left(\mathbb{E}(\sum_{\tau=t}^{\infty} \rho^{\tau-t} R_{j\tau}) - p_{jt} \right) \tag{7}$$

A4.3.3 Producer surplus

Per-period producer surplus in the housing unit market is:

$$\begin{aligned} PS_t(\{S\}_j;\theta) =& \mathbb{E}\left(\pi^*_{jt}\right) \\ =& Pr(\exp(\tilde{\xi}_{jt})\tilde{\eta} p_{jt}(0)^{\eta+1} - p_{jt} H_{jt-1} D > 0) \times \\ & \mathbb{E}(\exp(\tilde{\xi}_{jt})\tilde{\eta} p_{jt}(0)^{\eta+1} - p_{jt} H_{jt-1} D | \exp(\tilde{\xi}_{jt})\tilde{\eta} p_{jt}(0)^{\eta+1} - p_{jt} H_{jt-1} D > 0) \end{aligned}$$

with expectations defined over cost shocks. The firm earns positive profits for:

$$\begin{aligned} &\tilde{\xi}_{jt} > -\left(\log(\tilde{\eta}) + (\eta+1)\log(p_{jt}(0)) - \log(p_{jt}) - \log(H_{jt-1}) - \log(D)\right) \iff \\ &\tilde{\xi}_{jt} > -\left(\log(\tilde{\eta}) + \eta \log(p_{jt}) - \gamma(1+\eta) S_{jt} - \log(H_{jt-1}) - \log(D)\right) := -\tilde{\pi}_{jt} \\ &\quad \iff \xi^*_{jt} > \exp(-\tilde{\pi}_{jt}), \qquad \xi^*_{jt} = \exp(\tilde{\xi}_{jt}) \end{aligned}$$

So the expected value from the firm's perspective, who takes prices as given, is:

$$\begin{aligned} \mathbb{E}(\pi^*_{jt}) &= Pr(\xi^*_{jt} > \exp(-\tilde{\pi}_{jt})) \frac{\tilde{\eta} p_{jt}(0)^{\eta+1} \int_{\exp(\tilde{\pi}_{jt})}^{\infty} \xi^*_{jt} f(\xi^*_{jt}) d\xi^*_{jt} - p_{jt} H_{jt-1} D}{Pr(\xi^*_{jt} > \exp(-\tilde{\pi}_{jt}))} \\ &= \tilde{\eta} p_{jt}(0)^{\eta+1} \int_{\exp(-\tilde{\pi}_{jt})}^{\infty} \xi^*_{jt} f(\xi^*_{jt}) d\xi^*_{jt} - p_{jt} H_{jt-1} D Pr\left(\xi^*_{jt} > \exp(-\tilde{\pi})\right) \end{aligned}$$

If $\tilde{\xi}_{jt} \sim N(\mu, \sigma^2)$, then ξ^*_{jt} has a log-normal distribution and the incomplete integral from above (removing indices for clarity) is:

$$\begin{aligned} \int_{\exp(-\tilde{\pi})}^{\infty} \xi^* f(\xi^*) d\xi^* &= \int_{\exp(-\tilde{\pi})}^{\infty} \exp(\tilde{\xi}) f(\xi^*) d\xi^* \\ &= \int_{\exp(-\tilde{\pi})}^{\infty} \frac{\exp(\tilde{\xi})}{\xi^* \sigma \sqrt{2\pi}} \exp\left(-\frac{(\log(\xi^*) - \mu)^2}{2\sigma^2} \right) d\xi^* \end{aligned}$$

Let $\tilde{\xi} = \log(\xi^*) \Rightarrow d\tilde{\xi} = 1/\xi^* d\xi^*$ and $\xi^* = \exp(\tilde{\pi}) \Rightarrow \tilde{\xi} = \tilde{\pi}$. Then:

$$\begin{aligned}
\int_{\exp(-\tilde{\pi})}^{\infty} \xi^* f(\xi^*) d\xi^* &= \int_{-\tilde{\pi}}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp(\tilde{\xi}) \exp\left(-\frac{(\tilde{\xi}-\mu)^2}{2\sigma^2}\right) d\tilde{\xi} \\
&= \int_{-\tilde{\pi}}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(\tilde{\xi}^2 - 2\tilde{\xi}\mu + \mu^2 - 2\sigma^2\tilde{\xi})\right) d\tilde{\xi} \\
&= \int_{-\tilde{\pi}}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}(\tilde{\xi}^2 - 2\tilde{\xi}(\mu+\sigma^2) + \mu^2 + 2\mu\sigma^2 + \sigma^4 - 2\mu\sigma^2 - \sigma^4)\right) d\tilde{\xi} \\
&= \int_{-\tilde{\pi}}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\tilde{\xi}-(\mu+\sigma^2))^2}{2\sigma^2} + \mu + \sigma^2/2\right) d\tilde{\xi} \\
&= \exp(\mu+\sigma^2/2) \int_{-\tilde{\pi}}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\tilde{\xi}-(\mu+\sigma^2))^2}{2\sigma^2}\right) d\tilde{\xi} \\
&= \exp(\mu+\sigma^2/2)\left(1 - \Phi\left(\frac{-\tilde{\pi}-(\mu+\sigma^2)}{\sigma}\right)\right) \\
&= \exp(\mu+\sigma^2/2)\Phi\left(\frac{\mu+\sigma^2+\tilde{\pi}}{\sigma}\right)
\end{aligned}$$

Giving:

$$\mathbb{E}(\pi^*_{jt}) = \tilde{\eta} p_{jt}(0)^{1+\eta} \exp(\mu+\sigma^2/2)\Phi\left(\frac{\mu+\sigma^2+\tilde{\pi}_{jt}}{\sigma}\right) - p_{jt} H_{jt-1} D\Phi\left(\frac{\mu+\tilde{\pi}_{jt}}{\sigma}\right) \tag{8}$$

A4.3.4 Expected housing on each plot

The expected housing supply on each plot conditional on conditions in t (the relevant expectation for home prices) is:

$$\begin{aligned}
\mathbb{E}(H_{jt}|H_{jt-1}, p_{jt}) &= Pr(d_{jt}=0)H_{jt-1} + Pr(d_{jt}=1)\tilde{\eta} p_{jt}(0)^{\eta} \mathbb{E}(\xi^*_{jt}|d_{jt}=1) \\
&= Pr(\tilde{\xi}_{jt} < -\tilde{\pi}_{jt}) H_{jt-1} + \tilde{\eta} p_{jt}(0)^{\eta} \exp(\mu+\sigma^2/2)\Phi\left(\frac{\tilde{\pi}_{jt}+\mu+\sigma^2}{\sigma}\right) \\
&= H_{jt-1}\Phi\left(\frac{-\tilde{\pi}_{jt}-\mu}{\sigma}\right) + \mathbb{E}(H_{jt})\Phi\left(\frac{\tilde{\pi}_{jt}+\mu+\sigma^2}{\sigma}\right) \\
&= H_{jt-1}\left(1 - \Phi\left(\frac{\tilde{\pi}_{jt}+\mu}{\sigma}\right)\right) + \mathbb{E}(H_{jt})\Phi\left(\frac{\tilde{\pi}_{jt}+\mu+\sigma^2}{\sigma}\right) \\
&= H_{jt-1} - \Phi\left(\frac{\tilde{\pi}_{jt}+\mu}{\sigma}\right) H_{jt-1} + \mathbb{E}(H_{jt})\Phi\left(\frac{\tilde{\pi}_{jt}+\mu+\sigma^2}{\sigma}\right)
\end{aligned}$$

When $H_{jt-1} \leq \mathbb{E}(H_{jt})$ this is greater than H_{jt-1}: housing supply tends to increase, as development occurs when cost shocks are good. For values of $H_{jt-1} >> \mathbb{E}(H_{jt})$, it is possible that future housing supply can be lower than previous housing supply, conditional on prices.

A4.4 Propositions about changes in consumer, producer surplus

Consider the counterfactuals defined as follows:

C1 All future expected sinking is zero: $NPVS_{jt} := \sum_{\tau=t+1}^{\infty} \rho^{\tau-t} \mathbb{E}_t(s_{j\tau}) = 0 \, \forall j, t$.

C2 Mexico City experiences no sinking ever. That is, $NPVS_{jt} = S_{jt} = 0 \, \forall j, t$.

C3 Information frictions are resolved: $\theta \to 1$.

We evaluate each derivative "in the long run", where the long-run is long enough that housing on a plot has a chance to be redeveloped and update to a new equilibrium. In the long-run, given a demand shock (as all of our counterfactuals represent), the new equilibrium represents a movement along the supply curve, and we know the elasticity of this supply curve is η:

$$\frac{\partial \log(H_{jt})}{\partial \log(p_{jt}(0))} = \eta \Rightarrow \frac{\partial \log(H_{jt})}{\partial X_{jt}} = \eta \frac{\partial \log(p_{jt}(0))}{\partial X_{jt}}$$

for some demand shifter X_{jt}.

A4.4.1 Change in the probability of redevelopment

The probability that a plot is developed is:

$$Pr(\text{Redev}_{jt}) = \Phi\left(\frac{\mu + \tilde{\pi}_{jt}}{\sigma}\right), \qquad \tilde{\pi}_{jt} = \log(\tilde{\eta}) + \eta \log(p_{jt}(0)) - \gamma S_{jt} - \log(H_{jt}) - \log(D)$$

In counterfactual **C1**:

$$\begin{aligned}\frac{\partial Pr(\text{Redev}_{jt})}{\partial NPVS_{jt}} &\propto \eta \frac{\partial \log(p_{jt}(0))}{\partial NPVS_{jt}} - \frac{\partial \log(H_{jt})}{\partial NPVS_{jt}} \\ &= \eta \frac{\partial \log(p_{jt}(0))}{\partial NPVS_{jt}} - \eta \frac{\partial \log(p_{jt}(0))}{\partial NPVS_{jt}} = 0\end{aligned}$$

In counterfactual **C2**:

$$\frac{\partial Pr(\text{Redev}_{jt})}{\partial S_{jt}} \propto \frac{\partial Pr(\text{Redev}_{jt})}{\partial NPVS_{jt}} \frac{\partial NPVS_{jt}}{\partial S_{jt}} - \gamma = -\gamma > 0$$

In counterfactual **C3**:

$$\frac{\partial Pr(\text{Redev}_{jt})}{\partial \theta} \propto \eta \frac{\partial \log(p_{jt}(0))}{\partial \theta} - \frac{\partial \log(H_{jt})}{\partial \theta} = 0$$

A4.4.2 Change in producer surplus

Expected profits are:

$$PS_{jt} = \tilde{\eta} p_{jt}(0)^{\eta+1} \exp(\mu + \sigma^2/2) \Phi\left(\frac{\mu + \sigma^2 + \tilde{\pi}_{jt}}{\sigma}\right) - \exp(\gamma S_{jt}) p_{jt}(0) H_{jt} D \Phi\left(\frac{\mu + \tilde{\pi}_{jt}}{\sigma}\right)$$

Let $\Xi(\tilde{\pi}_{jt}) = \exp(\mu + \sigma^2/2)\Phi\left(\frac{\mu+\sigma^2+\tilde{\pi}_{jt}}{\sigma}\right)$ in the following, and recall that:

$$\frac{\partial \Xi(\tilde{\pi}_{jt})}{\partial NPVS_{jt}} = \frac{\partial \Xi(\tilde{\pi}_{jt})}{\partial \theta} = 0$$

In counterfactual **C1**:

$$\begin{aligned}
\frac{\partial PS_{jt}}{\partial NPVS_{jt}} &= \Xi(\tilde{\pi}_{jt})\tilde{\eta}(\eta+1)p_{jt}(0)^{\eta}\frac{\partial p_{jt}(0)}{\partial NPVS_{jt}} \\
&\quad - \exp(\gamma S_{jt})D\Phi\left(\frac{\mu+\tilde{\pi}_{jt}}{\sigma}\right)\left(\frac{\partial p_{jt}(0)}{\partial NPVS_{jt}}H_{jt} + \frac{\partial H_{jt}}{\partial NPVS_{jt}}p_{jt}(0)\right) \\
&= \Xi(\tilde{\pi}_{jt})\tilde{\eta}(\eta+1)p_{jt}(0)^{\eta}\frac{\partial p_{jt}(0)}{\partial NPVS_{jt}} \\
&\quad - \exp(\gamma S_{jt})D\Phi\left(\frac{\mu+\tilde{\pi}_{jt}}{\sigma}\right)\left(\frac{\partial p_{jt}(0)}{\partial NPVS_{jt}}H_{jt} + \eta\frac{\partial p_{jt}(0)}{\partial NPVS_{jt}}H_{jt}\right) \\
&= \frac{\eta+1}{p_{jt}(0)}\frac{\partial p_{jt}(0)}{\partial NPVS_{jt}}\left(\Xi(\tilde{\pi}_{jt})\tilde{\eta}p_{jt}(0)^{\eta+1} - p_{jt}H_{jt}D\Phi\left(\frac{\mu+\tilde{\pi}_{jt}}{\sigma}\right)\right) \\
&= (\eta+1)\frac{\partial \log(p_{jt}(0))}{\partial NPVS_{jt}}PS_{jt} = (\eta+1)\gamma\theta PS_{jt} \leq 0
\end{aligned}$$

In counterfactual **C2**:
The change in producer surplus is equal to that of **C1**, plus the change due to the change in S_{jt}:

$$\begin{aligned}
\frac{\partial PS_{jt}}{\partial S_{jt}} = &\underbrace{\frac{\partial PS_{jt}}{\partial NPVS_{jt}}\frac{\partial NPVS_{jt}}{\partial S_{jt}}}_{\leq 0}\underbrace{-\gamma p_{jt}H_{jt}D\Phi\left(\frac{\mu+\tilde{\pi}_{jt}}{\sigma}\right)}_{>0} \\
&\underbrace{-\frac{\gamma}{\sigma}\left(\tilde{\eta}p_{jt}(0)^{\eta+1}\exp(\mu+\sigma^2/2)\phi\left(\frac{\mu+\sigma^2+\tilde{\pi}}{\sigma}\right) - p_{jt}H_{jt}D\phi\left(\frac{\mu+\tilde{\pi}}{\sigma}\right)\right)}_{\lesseqgtr 0}
\end{aligned}$$

The sign of this derivative is unknown. However, some patterns emerge:

- As very high or low $\tilde{\pi}_{jt}$, the last term goes to zero, and there is little impact through the changed probability of re-development.
- For values of $\tilde{\pi}_{jt}$ near the middle of the distribution, where cost shocks are more marginal to the decision to develop, things are more complicated for this term.
- Sending this term to zero, we see that benefits are not clear to producers. Subsidence benefits them by lowering the cost of development (term 2), but it also lowers revenues (term 3). As $\theta \to 0$, the costs channel wins.

In **C3**:

$$\begin{aligned}\frac{\partial PS_{jt}}{\partial \theta} =& \frac{\partial \tilde{\pi}_{jt}}{\partial \theta}\left(\tilde{\eta} p_{jt}(0)^{\eta+1}\exp(\mu+\sigma^2/2)\phi\left(\frac{\mu+\sigma^2+\tilde{\pi}}{\sigma}\right) - p_{jt}H_{jt}D\phi\left(\frac{\mu+\tilde{\pi}}{\sigma}\right)\right) \\ &+ \frac{\partial \log(p_{jt}(0))}{\partial \theta}(\eta+1)PS_{jt} \\ =& \frac{\partial \log(p_{jt}(0))}{\partial \theta}(\eta+1)PS_{jt} \leq 0\end{aligned}$$

Information frictions unambiguously increase producer surplus by increasing revenue.

A4.5 Proof of validity of control function

The full structural supply model is:

$$\begin{aligned} d_{jt} &= 1\left(\tilde{V}^S_{jt} \geq 0\right) \\ \tilde{V}^S_{jt} &= \log(\tilde{\eta}) + \eta\log(p_{jt}) - \gamma(1+\eta)S_{jt} - \log(H_{jt-1}) - \log(D) + \log(\xi_{jt}) \\ \log(p_{jt}) &= \gamma S_{jt} + \Pi\text{Rel. altitude change}_{jt} + \tilde{u}_{jt}\end{aligned}$$

The proof follows Blundell and Matzkin (2014), in showing that the residual $\tilde{u}_{jt}$ satisfies control function separability, which follows naturally from the linearity of the first stage:

$$r^2(\tilde{V}^S_{jt}, \log(p_{jt}), Z_{jt}) = \tilde{u}_{jt} = \underbrace{\log(p_{jt}) - \gamma S_{jt} - \Pi\text{Rel. altitude change}_{jt}}_{q(\log(p_{jt}), Z_{jt})}$$

Note that $v(q(\log(p_{jt}), \text{Rel. altitude change}_{jt}), r^1(\log(p_{jt}))) = q(\log(p_{jt}), \text{Rel. altitude change}_{jt})$, so q is strictly increasing in $\overline{p}_{jt}$ and $v()$ is strictly increasing in $q()$. Thus, the structural inverse of the system satisfied control function separability, and a control function can be derived from the first stage to estimate the second stage.

A4.6 Computing counterfactuals

Given our sparse data on prices, we seek to estimate counterfactual prices and housing densities using a dataset of plots as a single cross-section. To do this, we consider a steady state in which the joint distribution of the underlying random variables that generate the equilibrium, $(\xi_{jt}, \varepsilon^i_{jt}, e_{jt}, s_{jt})$ is time-invarying and approximately non-stochastic given the large number of plots we consider (985,862). The intution of this steady state is that while the draws of cost and preference shocks are stochastic, given enough plots the distribution of $(p_{jt}(0), H_{jt})$ is constant in each period and can be treated as fixed. In the steady state, we drop the t subscript to emphasize the independence of time.

The equilibrium is most easily expressed in terms of log prices and log relative housing relative to the outside option. Let $\tilde{\lambda}_j = \log(\lambda_j) - \log(\lambda_0)$; in equilibrium, $\tilde{\lambda}_j = \log(H_j) -$

$\log(H_0)$. The equilibrium is characterized by:

$$\tilde{\lambda}_j^D = -\log(R_j(0)) + Z_j\delta + e_j \tag{9}$$

$$\tilde{\lambda}_j^S = \eta \log\left(\frac{\eta}{1+\eta}\right) + \eta \log(p_j(0)) + \tilde{\xi}_j - \log(H_{0t}^S) \tag{10}$$

$$\tilde{\lambda}_j^D \approx -\log(p_{jt}(0)) + Z_j\delta + e_j + \gamma\theta \sum_{\tau=t}^{\infty} \rho^{\tau-t} \mathbb{E}_t(s_{j\tau}) \tag{11}$$

The last equation is derived directly from Equation (14) evaluated at $S_{jt} = 0$.

Given data on the plot level on $(\tilde{\lambda}_j, \log(p_j(0)), Z_j, \sum_{\tau=t}^{\infty} \rho^{\tau-t}\mathbb{E}_t(s_{j\tau}))$ and estimates or calibrated values of the parameters $(\gamma, \theta, \delta, \eta, \rho)$, we first solve the previous equations exactly to back out $(e_j, \tilde{\xi}_j)$. From the inferred values of $\tilde{\xi}_j$, we calibrate μ and σ as the mean and standard deviation of these shocks.

In order to calculate welfare under different counterfactuals, for example setting $\mathbb{E}_t(s_{j\tau}) = 0$ or $\theta = 1$, we alter these parameters and, holding $(Z_j, e_j, \tilde{\xi}_j)$ fixed, solve the system for alternative vectors of prices and relative housing supply that clear the housing market on each plot.

A4.7 Proof that the new error function is independent of prices

In order for the control function approach to provide unbiased estimation of the coefficient on prices, prices and the (modified) error term must be independent. We prove that here:

$$\begin{aligned}
\mathbb{E}\left(\tilde{\xi}_{jt}^* | \log(p_{jt})\right) &= \mathbb{E}\left(\mathbb{E}\left(\tilde{\xi}_{jt}^* | \log(p_{jt}), Z_{jt}\right) | \log(p_{jt})\right) \\
&= \mathbb{E}\left(\mathbb{E}\left(\tilde{\xi}_{jt} - \mathbb{E}(\tilde{\xi}_{jt}|\tilde{u}_{jt}) | \log(p_{jt}), Z_{jt}\right) | \log(p_{jt})\right) \\
&= \mathbb{E}\left(\mathbb{E}\left(\tilde{\xi}_{jt} | \log(p_{jt}), Z_{jt}\right) - \mathbb{E}\left(\mathbb{E}(\tilde{\xi}_{jt}|\tilde{u}_{jt}) | \log(p_{jt}), Z_{jt}\right) | \log(p_{jt})\right) \\
&= \mathbb{E}\left(\mathbb{E}\left(\tilde{\xi}_{jt} | \log(p_{jt}), Z_{jt}\right) - \mathbb{E}\left(\mathbb{E}(\tilde{\xi}_{jt}|\tilde{u}_{jt}) | \tilde{u}_{jt}, Z_{jt}\right) | \log(p_{jt})\right) \\
&= \mathbb{E}\left(\mathbb{E}\left(\tilde{\xi}_{jt} | \tilde{u}_{jt}, Z_{jt}\right) - \mathbb{E}\left(\tilde{\xi}_{jt} | \tilde{u}_{jt}\right) | \log(p_{jt})\right) \\
&= \mathbb{E}\left(\mathbb{E}\left(\tilde{\xi}_{jt} | \tilde{u}_{jt}\right) - \mathbb{E}\left(\tilde{\xi}_{jt} | \tilde{u}_{jt}\right) | \log(p_{jt})\right) = \mathbb{E}\left(0 | \log(p_{jt})\right) = 0
\end{aligned}$$

A5 Cost-benefit Appendix

A5.1 Pigouvian tax

Our empirical estimates provide a measure of the total cost of subsidence, which expressed in terms of dollars per meter of average elevation loss is:

$$\frac{\text{Total cost}}{\text{meter of elevation loss}} = \frac{\text{Total cost}}{\overline{NPVS}} = \frac{1}{0.931}\text{Total cost} \tag{12}$$

In order to price the externality-generating action, groundwater pumping, we must convert this to dollars per unit of water pumped.

Subsidence is caused by low groundwater tables. We estimate the response of subsidence rates to changes in aquifer storage by taking the average change in subsidence rates over the 13-year period in our panel, and estimating the dosage-response of subsidence rates to changes in aquifer storage:

$$\begin{aligned} \Delta s &= \frac{\Delta s}{\Delta V} \times \Delta V \\ \Delta s &= \underbrace{\frac{0.0361}{-111 \times 13}}_{\substack{\text{Annual loss} \times T = \\ \text{Total loss} \\ \text{over 13 years}}} \Delta V \end{aligned} \tag{13}$$

Here Δs is the change in the annual subsidence rate (change in $m/$ year), and ΔV is the net loss in aquifer storage (millions of cubic meters, or hectometers hm^3). We take the annual net loss in aquifer storage from CONAGUA (2024); details from this report on aquifer demand over this period are described in Table AA3.[4]

This is how a change in volume translates to a change in the sinking rate; finally, to get this in terms of the present value of the change in storage, we must consider the discounted stream of offset sinking from changes in volume each year. Thus, the Pigouvian tax on water consumption takes the form:

$$\tau = \frac{1}{0.931}\text{Total cost} \times \frac{\Delta s}{(1-\rho)\Delta V} \tag{14}$$

A5.2 Cost-benefit of other recharge policies

Here we discuss the costs and benefits of these policies, a challenging task given the lack of detailed information on the cost structure of some of these interventions. We discuss how we construct the costs and benefits of each of the explored policies, with emphasis on the limitations of these estimates.

[4]This expression extracts away from dynamics; how long it would take for a higher water table to translate to slower sinking rates is difficult to answer but crucial question for policy. While it may take decades for higher aquifer levels to translate to fully abated sinking, interventions that locally rise water tables such as wastewater injection sites have seen rapid abatement of subsidence.

Table AA3: Aquifer demand

	Annual volume (hm^3)
Natural recharge	151.4
Recharge from drinking water leaks	361.4
Total recharge	512.8
Pumping for drinking water	448.5
Other pumping	175.3
Total extraction	623.8
Net volume change per year	-111

Notes: Source: CONAGUA (2024).

The net benefits of a policy that facilitates aquifer recharge is the gains from offsetting subsidence minus the sum of its treatment, procurement and construction costs net what the current system costs, capping potential benefits at the total recharge needed to stop subsidence. We make three simplifying assumptions. First, under the status quo the average subsidence rate remains constant at $\bar{s} = 2.5$ cm/year; we abstract from further acceleration due to the ongoing net storage, which makes our benefit estimates conservative. Second, water added to the aquifer remains in storage indefinitely and continues to reduce subsidence after an intervention ends. Third, the relationship $\Delta s/\Delta V$ is linear down to zero subsidence. Benefits are therefore bounded: once an intervention has raised storage by $\bar{S} = \bar{s}/(\partial s/\partial V) \approx 999$ hm^3, average subsidence is fully abated and additional water yields no further benefit.

$$\frac{\text{Benefits}}{\text{Costs}} = \frac{\sum_{t=0}^{T} \rho^t \, \tau \, \min\Big(\sum_{k \leq \tau} q_k, \; \bar{S} \Big)}{\sum_{\tau=0}^{T} \rho^\tau (\text{Treatment}_\tau + \text{Operating}_\tau + \text{Construction}_\tau)}, \qquad \bar{S} = \frac{\bar{s}}{\partial s / \partial V}$$

Treatment costs are the costs of treating the water, which is a function of the source water quality (aquifer, wastewater, surface water) and its intended use (drinking, injection). *Operating costs* are the costs of bringing the water from its location (aquifer, surface source) into the Mexico City water supply or running the program (in the case of fixing leaks). *Construction* costs are the costs of building the proposed investment. q_k is the water saved in year k.

For all calculations, USD costs are deflated with the US CPI-U and peso costs with the Mexican CPI (INPC), then converted at 19 MXN/USD (2018). Per-m^3 costs apply to the volume saved in each year.

A5.2.1 Wastewater injection

We take costs of investing in wastewater injection and the potential capacity of an injection site from López-Morales and Mesa-Jurado (2017), who characterize the costs of treating wastewater and injecting in the Chichinautzin Sierra.

Table AA4: Values and sources for wastewater injection costs

Cost type	Explanation and source
Treatment costs	Cost of treating water to be clean enough to enter the aquifer without activating heavy metals or causing well clogs.
Operating costs	Annual costs of operation, treatment
Construction costs	Building treatment and injection infrastructure.

A5.2.2 Leaks control

Table AA5: Leaks repair costs

Cost type	Explanation
Treatment costs	There is no change in treatment costs when fixing leaks because the water supply mix is unaffected.
Operating costs	None; all modeled as initial investment.
Construction costs	$163.3M for sectorization, plus two detection programs of $12.95M each (years 0-12 and 13-25).

We take reports of the costs and efficiency of efforts to repair leaks in the system from López-Morales and Mesa-Jurado (2017). Repairing leaks may have contradictory impacts on the aquifer; CONAGUA (2024) estimate that most recharge on the aquifer comes from lost water in the Mexico City system, and considering that some of that water is imported from outside the watershed would imply that some of this recharge is additional to what is pumped. We take an optimistic scenario and assume that all leaks that are addressed are done on parts of the system *prior* to entering the city, which would make all water saved additional. However, given that our cost estimates are calculated using efforts conducted *within* the city, it is unclear whether we over- or under-estimate these relative to the true costs of repairing leaks outside the city. While infrastructure outside the city is exposed and therefore less costly to work on than buried lines, these lines are also larger and more economically costly to shut down.

A5.2.3 New water sources

We take reports of the costs and efficiency of efforts to generate new water sources from López-Morales and Mesa-Jurado (2017). They consider five alternative sources of water: Temascaltepec, a surface water system that connects with the Cutzamala System; Tecolutla (Veracruz); Amacuzac (Morelos); Taxhimay, a dam in the Estado de México; and Oriental-Libres, a groundwater basin in the state of Puebla. Our final estimate for alternative water sources averages across these. We use their operating costs and initial investment estimates, but assume that construction takes 6 years and use our own discount rate to aggregate costs.

Table AA6: New water sources costs

Cost type	Explanation
Treatment costs	We assume no change in treatment costs. These are potentially even net negative, as it may be cheaper to treat surface water than groundwater which is heavily contaminated with Arsenic.
Operating costs	Conveyance and storage costs.
Construction costs	Infrastructure investment to transport surface water to the city.

A5.2.4 Wastewater recycling

Table AA7: Wastewater recycling costs

Cost type	Explanation
Treatment costs	This is difficult to characterize; while grey water is very costly to treat for reintroduction into the water supply, the high Arsenic content of the groundwater currently being pumped is also costly to treat.
Operating costs	Electricity, chemical, and personnel required to run the plant.
Construction costs	Infrastructure investment to build plant and accompanying infrastructure.

Our primary source for information the costs and benefits of investing in wastewater recycling plants comes from the 2007-2012 Plan for Sustainable Water Management in Mexico City (Gobierno del Distrito Federal, 2007). This document describes an investment in building 6 macro-sites for secondary wastewater treatment. We assume a useful life of 50 years once built following engineering guidance. In the absence of information on operating and treatment costs *relative* to the current cost of treating water from the aquifer, which this would offset, we use a value of $0.34 USD per cubic meter, the same cost as treatment for wastewater injection. If we take the cost of treatment for wastewater injection as the benchmark cost of treating water that comes out of the aquifer, this would imply that treating wastewater is twice as expensive as treating groundwater. To the extent that treating groundwater is more expensive than treating water for injection, this margin is smaller. Also note that construction costs may have evolved substantially between 2007 and now, and better estimates of operating costs are necessary for comprehensive analysis.